

POINT OPERATIONS AND MATCHING-LINE BUBBLES IN THE ATIYAH–FLOER COMPARISON

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ABSTRACT. Daemi, Fukaya, and Lipyanskiy proved the admissible $SO(3)$ Atiyah–Floer comparison and compatibility of the comparison map with the operations associated with H_1 and H_2 of the splitting surface. The point class is the remaining generator; its comparison homotopy first encounters energy concentration on the matching line, the codimension-one seam between the gauge-theoretic and symplectic regions, in Fredholm index four.

We compactify the one-dimensional point-cut mixed moduli space and classify its new boundary. Chern–Simons excision gives positive half-integral local charge, which at index four forces a single charge-one matching bubble. Relative stable reduction identifies the primitive limits, a Dirichlet Hermitian–Yang–Mills problem reconstructs them, and a scale-uniform inverse gives converse gluing. Determinant-line excision reduces the oriented local count to Muñoz’s extension spaces.

The resulting boundary identity shows that uncorrected point-operation compatibility fails in genera one and two. After dividing the positive Pontryagin point class by four, the symplectic point operation must be shifted by 2, 1, or 0 when the marked point lies on a component of genus one, two, or at least three; in the integral p_1 normalization the corresponding local counts are 8, 4, and 0. Together with the surface and loop results of Daemi, Fukaya, and Lipyanskiy, this proves the full corrected observable-algebra action in the admissible embedded-Lagrangian setting.

1. INTRODUCTION

1.1. The point operation. Let $(Y_\#, E_\#)$ be an admissible $SO(3)$ pair split along a closed oriented surface Σ . Atiyah’s gauge–symplectic comparison principle predicts an identification of the two resulting Floer theories [Ati88]. In the embedded-Lagrangian setting of Daemi–Fukaya–Lipyanskiy, henceforth DFL, the splitting produces an instanton Floer complex C_G with differential d_G , a Lagrangian Floer complex C_S with differential d_S , and a comparison chain map

$$N : C_G \longrightarrow C_S.$$

Their main theorem identifies the induced homologies [DFL26, Theorem 4]. The comparison is defined by counting solutions of a mixed equation: an anti-self-dual connection on one side of a codimension-one seam and a pseudoholomorphic map on the other, with their boundary values identified in the symplectic moduli space of flat connections [AB83]. The regularity, Fredholm, compactness, and removal theorems for this equation are proved in [DFL24].

The universal bundle over the flat moduli space and the splitting surface gives operations indexed by $H_*(\Sigma)$. DFL prove that N intertwines the operations associated with $H_2(\Sigma)$ and $H_1(\Sigma)$ [DFL26, Section 7.2]. They also explain why the same argument stops at $H_0(\Sigma)$: a

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point insertion requires four-dimensional mixed moduli spaces, and these can bubble on the matching line. A neighborhood of that boundary was not part of their compactification.

The purpose of this paper is to isolate the obstruction and construct the compactification needed to describe and count that neighborhood. The cylinder calculation shows that uncorrected compatibility is impossible in low genus. The general formula attributes the discrepancy to a matching-line scalar depending only on the genus of the component carrying the point.

The local charge, primitive stable reduction, Dirichlet reconstruction, uniform inverse, no-neck theorem, primitive gluing collar, and stratified point incidence are established below. Determinant excision identifies the local algebraic multiplicities and cylinder calibration fixes their coherent signs. The resulting oriented boundary formula proves the point comparison and its full-operation consequence.

Convention 1.1. All moduli spaces use DFL's orientation, grading, and end conventions. The symbol p_1 denotes the positive integral degeneracy cycle determined by their product orientation. Unless an integral statement is explicitly made, coefficients are rational. A *matching atom* is an atom of the mixed energy measure at a point of the matching line. It is *primitive* when its relative clutching number is one, equivalently when its normalized energy is $\frac{1}{2}$. Normalized energy means DFL's topological energy, recalled in (3).

1.2. **Results.** Write

$$\Sigma = \bigsqcup_j \Sigma_j, \quad g_j = \text{genus}(\Sigma_j).$$

We retain all admissibility, irreducibility, and embeddedness hypotheses of DFL. In particular, the standard secondary perturbations are supported away from the matching line. For $q \in \Sigma_j$, let b_q^G and b_q^S be the gauge and symplectic operations represented by the positive integral degeneracy cycle $p_1/[q]$. Over $\mathbb{Q}$ put

$$m_q^G = \frac{1}{4}b_q^G, \quad m_q^S = \frac{1}{4}b_q^S.$$

This is DFL's positive quarter-Pontryagin convention. The sign is fixed by their product orientation on the linear-dependence cycle, rather than by the alternative convention $\mu(q) = -\frac{1}{4}p_1/[q]$ sometimes used in instanton theory [DK90]. For the cylinder splitting of $\Sigma_g \times S^1$, abbreviate the corresponding point operations by b_G and b_S .

Theorem A (low-genus cylinder obstruction; = Proposition 5.3). *For the cylinder splitting of $\Sigma_g \times S^1$, the positive integral point operations satisfy*

$$Nb_G - b_S N = \begin{cases} 8N, & g = 1, \\ 4N, & g = 2. \end{cases}$$

Consequently, the uncorrected point comparison cannot hold in general in genus one or two.

Theorem A is proved independently of the matching-line compactification: it follows from DFL's surface-operation comparison and Muñoz's cylinder relations. Proposition 5.3 gives the proof.

Theorem B (point-class comparison; = Theorem 7.7). *There are compatible secondary perturbations and a point-cut chain homotopy $K_q : C_G \rightarrow C_S$ for which*

$$(1) \quad Nm_q^G - m_q^S N = d_S K_q + K_q d_G + \kappa_{g_j} N, \quad \kappa_g = \begin{cases} 2, & g = 1, \\ 1, & g = 2, \\ 0, & g \geq 3. \end{cases}$$

Equivalently, N intertwines the gauge point operation with the corrected symplectic operation

$$\widehat{m}_q^S = m_q^S + \kappa_{g_j} \text{id}$$

up to chain homotopy.

Theorem B is restated as Theorem 7.7; its integral normalization is Theorem 8.1.

The integral version of (1) has coefficients 8, 4, and 0; see Theorem 8.1. We state the quarter-Pontryagin formula over $\mathbb{Q}$ because an integral refinement requires a separate divisibility choice for the universal bundle.

Theorem B completes the list of generator comparisons. Give $H_k(\Sigma; \mathbb{Z})$ degree $4 - k$, and let $\mathbb{A}_{p_1}(\Sigma; R)$ denote the free graded-commutative algebra on $H_*(\Sigma; \mathbb{Z}) \otimes_{\mathbb{Z}} R$.

Corollary C (full corrected operation algebra; = Corollary 8.4). *For every commutative ring R , scalar extension of the DFL comparison is an isomorphism of $\mathbb{A}_{p_1}(\Sigma; R)$ -modules after the symplectic point generator is corrected by 8, 4, or 0 in genera one, two, or at least three. If 4 is invertible in R , the quarter-Pontryagin correction is respectively 2, 1, or 0. The intertwining identity for every chosen word is induced by an explicit signed chain homotopy.*

Corollary C is the formal generator-by-generator consequence proved in Corollary 8.4. Its only new input beyond DFL is Theorem B.

The scalar term comes from the following compactification theorem. Let M_g denote the moduli space of flat connections on the nontrivial $SO(3)$ bundle over a connected genus- g surface. Under the Narasimhan–Seshadri correspondence it is the moduli space of stable rank-two bundles with fixed odd determinant [NS65]. Muñoz’s extension space and compact primitive bundle space are defined as follows. Fix the determinant line Λ , and let $\mathcal{E}_g$ be the rank- g vector bundle on $J(\Sigma_g)$ with fiber $H^1(\Sigma_g, L^2 \otimes \Lambda^{-1})$ over L . Then

$$(2) \quad N_g = \mathbb{P}(\mathcal{E}_g^\vee) \longrightarrow J(\Sigma_g), \quad \mathfrak{M}_g = \mathbb{P}((\mathcal{E}_g \oplus \mathcal{E}_g)^\vee) \longrightarrow J(\Sigma_g),$$

and there is a natural map $i_g : N_g \rightarrow M_g$ [Muñ99b]. A point of a fiber of $\mathfrak{M}_g$ is a pair of extension vectors $[v_0 : v_1]$. Independent vectors give a minimal rational curve; dependent vectors give a degree-one quasimap with one basepoint.

For the next statement, a *primitive matching bubble* means a rescaled limit with relative clutching charge one. We call it *full rank* when v_0 and v_1 are independent and *rank drop* when they are dependent. An algebraic limit is *admissible along the seam* when every surface restriction over the compactified matching line is stable and any quasimap basepoint lies on the gauge-theoretic hemisphere.

Theorem D (primitive compactification and gluing). *Let a sequence in an index-four mixed moduli space develop a matching-line bubble while meeting the point incidence. Then its residual mixed solution has index zero and the following hold.*

- (1) *There is one matching concentration point, its relative clutching charge is one, its normalized energy is $\frac{1}{2}$, and every neck and smaller scale carries zero energy.*

- (2) *After passage to a subsequence, finite-scale holomorphic doubling and relative stable reduction produce a primitive algebraic limit in $\mathfrak{M}_g$. It is either a full-rank minimal line or a rank-drop bundle with one positive Hecke modification. Both attach to the residual solution through $i_g(N_g) \subset M_g$.*
- (3) *Every algebraic limit admissible along the seam is realized by an exact local mixed solution. The primitive stratum has a gluing collar, and the gluing map is a local diffeomorphism onto a neighborhood of the bubbled face.*
- (4) *After the integral p_1 point cut, the oriented local contribution per residual solution is*

$$c_g = \begin{cases} 8, & g = 1, \\ 4, & g = 2, \\ 0, & g \geq 3. \end{cases}$$

The atomwise charge statement in Theorem D and the uniqueness of the physical matching center follow from Theorem 3.3 and Corollary 3.4. Proposition 6.4 excludes analytic neck energy, and Theorems 6.2 and 6.10 reconstruct the primitive bubble and give its collar. Proposition 7.4 supplies the transverse compact point cut. Proposition 7.5 and Section 7.5 prove part (4).

The two types in part (2) are both necessary in the compactification. Convergence away from a physical concentration point fixes the rescaled evaluation at infinity; it does not force a rescaled map to be constant at finite points. The generic genus-two bubble is therefore full rank, while rank drop is a compactification stratum. In genus one every pair of extension vectors is dependent and only the rank-drop, purely gauge-theoretic branch remains.

1.3. Why the correction is forced. The low-genus constants can be seen before doing any matching-line analysis. Take the cylinder splitting of $\Sigma_g \times S^1$. On the gauge side the point operation is multiplication by Muñoz’s degree-four Floer generator b_G ; on the symplectic side it is quantum multiplication by the geometric generator b_S . For $g = 2$ the ring relations are

$$a_G^2 + b_G - 8 = 0, \quad a_S^2 + b_S - 4 = 0,$$

where a is the degree-two surface generator [Muñ99a, Muñ99b]. DFL already prove $Na_G = a_S N$, and hence

$$Nb_G - b_S N = 4N.$$

For $g = 1$, the odd-determinant character variety is a point, whereas the instanton Floer relation is $b_G = 8 \text{id}$; therefore

$$Nb_G - b_S N = 8N.$$

These identities use only the established surface compatibility and do not assume point compatibility. They prove that an uncorrected theorem is impossible and later calibrate the local determinant orientation.

1.4. Method. The point-cut homotopy is zero-dimensional in mixed index three. Its boundary relation is one-dimensional in mixed index four. DFL’s energy-index formula and half-integrality imply that any defective index-four sequence loses exactly $\frac{1}{2}$ unit of topological energy and has an index-zero residual solution. This global statement does not by itself prevent several nonquantized matching atoms.

The proof has four stages. First, on a small circle about a matching point, the gauge and symplectic halves are spliced to a connection on $S^1 \times \Sigma_g$. Comparing the original smooth filling with the removable limiting filling identifies their mixed Chern–Simons branch difference with $n/2$, where n is the Pontryagin number of a relative clutching bundle on $S^2 \times \Sigma_g$. The total

defect $1/2$ then forces $n = 1$ at one point. A good-circle Chern–Simons argument excludes energy on the intervening neck.

The second step avoids the usual adiabatic projection. After base dilation the vertical-curvature term in the ASD equation is multiplied by the square of the scale, rather than its inverse. The usual adiabatic projection to M_g therefore does not apply; compare [DS94, DS07]. Instead, every smooth finite-scale mixed solution determines a holomorphic bundle on the full disc times Σ . Relative stable reduction associates to these doubled bundles a classifying map and a finite modification cycle. Relative p_1 additivity gives

$$n = d + \ell.$$

The primitive identity $n = 1$ leaves exactly a minimal line $(d, \ell) = (1, 0)$ or one rank-drop modification $(d, \ell) = (0, 1)$.

The third stage is reconstruction and gluing. Decomposing $\mathbb{P}^1$ into two hemispheres and prescribing the Narasimhan–Seshadri seam metrics puts the gauge hemisphere in the setting of Donaldson’s Dirichlet theorem for Hermitian Yang–Mills metrics [Don92]. The Dirichlet solution reconstructs every seam-admissible algebraic point. The reduced Fredholm problem is the Dolbeault deformation complex, and an anisotropic difference-quotient estimate keeps its right inverse bounded as the gauge hemisphere collapses. Pregluing and contraction then give the primitive collar.

The fourth stage treats incidence and orientation. Stratified transversality extends the point cut over the collar. Determinant-line excision separates the residual orientation from a universal local scalar. The image of i_g has real codimension $2(g - 2)$, the genus-two map i_2 has degree four, and the genus-one excess line has degree eight. The Dolbeault determinant identifies these numbers with the matching-face counts, and the cylinder identities fix both coherent signs as positive.

1.5. Relation to prior work. The mixed equation and the admissible comparison theorem are due to DFL [DFL26, DFL24]. The earlier adiabatic comparison of instantons and holomorphic curves is due to Dostoglou–Salamon [DS94, DS07]. Wehrheim established energy quantization and removal for ASD instantons with Lagrangian boundary conditions [Weh05a] and the energy identity on $\mathbb{C} \times \Sigma$ [Weh06]. Owens proved the correspondence between finite-energy instantons on a cylinder and stable bundles on $\mathbb{P}^1 \times \Sigma$ [Owe01]; the stable-family-plus-modification description used here is its relative local form.

Muñoz computed both the instanton Floer ring of $\Sigma \times S^1$ and the quantum cohomology of M_g [Muñ99a, Muñ99b]. His projective-bundle description of primitive product bundles supplies the finite-dimensional compactification and the two low-genus numbers. The appearance of a correction term is analogous in spirit to seam-bubble corrections in pseudoholomorphic quilts [BW18], but the local object here couples an ASD half to a holomorphic half and is not identified with a figure-eight bubble.

2. THE POINT CUT IN THE MIXED MODULI SPACE

2.1. The DFL setting. Let

$$Y_{\#} = Y \cup_{\Sigma} (-Y')$$

be an admissible splitting in the sense of [DFL26, Sections 2–3]. The restriction F of the $SO(3)$ bundle to every component of Σ is nontrivial. We assume, as in the embedded form of the DFL theorem, that the perturbed three-manifold Lagrangians are embedded and intersect

transversely and that all flat connections in the construction are irreducible. Put

$$M = \mathcal{M}(\Sigma, F) = \prod_j M_j, \quad M_j = \mathcal{M}(\Sigma_j, F_j).$$

The standard complex structure J_* identifies M_j with the fixed-odd-determinant stable-bundle moduli space. Its real dimension is $6g_j - 6$ for $g_j \geq 2$; M_j is a point for $g_j = 1$.

For generators $\alpha \in C_G$ and $\beta \in C_S$, let

$$\mathcal{M}_\eta(\alpha, \beta)_r$$

denote the moduli space of perturbed mixed solutions whose linearized mixed operator has Fredholm index r . The secondary perturbation η is a finite sum of gauge-theoretic holonomy perturbations and domain-dependent almost-complex variations. It is identically zero on a fixed neighborhood of the matching line. DFL construct such perturbations regular through index three [DFL26, Proposition 3.23].

For a mixed pair (A, u) , their normalized topological energy is

$$(3) \quad \mathcal{E}(A, u) = \frac{1}{8\pi^2} \int_X \text{tr}(F_A \wedge F_A) + \frac{1}{4\pi^2} \int_S u^* \Omega,$$

with the perturbation terms incorporated on the cylindrical pieces. The index formula has the form

$$(4) \quad \text{ind } \mathcal{D}_{(A, u)} = 8\mathcal{E}(A, u) + \varepsilon(\beta) - \varepsilon(\alpha).$$

For two mixed pairs with the same ends,

$$(5) \quad 2(\mathcal{E}(A, u) - \mathcal{E}(A', u')) \in \mathbb{Z};$$

see [DFL26, Lemma 3.15]. These two formulas are the numerical source of the index-four threshold.

2.2. The universal degeneracy cycle. Let $\tilde{\mathbb{V}}$ be DFL's interpolating universal adjoint bundle over the mixed configuration space times the glued parameter space $\tilde{X}$. Its complexification has rank three. For two generic sections s_1, s_2 , let

$$\mathcal{D}(s_1, s_2) = \{x : \text{rank}_{\mathbb{C}}(s_1(x), s_2(x)) \leq 1\}.$$

At the top rank-one stratum this has complex codimension two and hence real codimension four. With DFL's product orientation it represents the positive class $p_1(\tilde{\mathbb{V}})$. The rank-zero stratum has larger codimension and is absent from every moduli space considered below.

The interpolation from the gauge insertion to the symplectic insertion is an embedding

$$\iota : \mathbb{R} \times \Sigma \longrightarrow \tilde{X}.$$

The parameter $t \in \mathbb{R}$ tends at the two ends to the representatives used to define b_q^G and b_q^S . If $P \subset \Sigma$ is an oriented smooth k -cycle, define

$$\mathcal{L}_d^P(\alpha, \beta) \subset \mathcal{M}_\eta(\alpha, \beta)_r \times \mathbb{R}_t \times P$$

by the condition

$$s_1([A, u], \iota(t, x)) \wedge s_2([A, u], \iota(t, x)) = 0.$$

Lemma 2.1 (dimension of the comparison cut). *If the mixed moduli space and the degeneracy cycle are transverse, then $\mathcal{L}_d^P(\alpha, \beta)$ has expected dimension d precisely when*

$$(6) \quad r = d + 3 - k.$$

Proof. The mixed moduli, the interpolation parameter, and the cycle P have total dimension $r+1+k$. The top degeneracy stratum has real codimension four. Thus the cut has dimension $r+k-3$, which equals d exactly when (6) holds. $\square$

For a point $q \in \Sigma$, the zero-dimensional comparison homotopy uses index three and the one-dimensional relation uses index four:

$$(7) \quad \begin{aligned} \mathcal{L}_0^q(\alpha, \beta) &\subset \mathcal{M}_\eta(\alpha, \beta)_3 \times \mathbb{R}_t, \\ \mathcal{L}_1^q(\alpha, \beta) &\subset \mathcal{M}_\eta(\alpha, \beta)_4 \times \mathbb{R}_t. \end{aligned}$$

This is the first insertion for which DFL's construction reaches the matching-line bubbling threshold.

2.3. The point-cut homotopy. By Proposition 3.1 below, mixed moduli spaces of index at most three do not bubble. Once transversality is imposed, the oriented zero-manifolds $\mathcal{L}_0^q(\alpha, \beta)$ therefore define

$$(8) \quad K_q^{p_1}(\alpha) = \sum_{\beta} \# \mathcal{L}_0^q(\alpha, \beta) \beta.$$

Set $K_q = \frac{1}{4} K_q^{p_1}$ over $\mathbb{Q}$.

The ordinary compactification of $\mathcal{L}_1^q$ has four familiar boundary types: gauge and symplectic Floer breaking and escape of t to its gauge and symplectic ends. With DFL's conventions their signed sum is

$$(9) \quad d_S K_q^{p_1} + K_q^{p_1} d_G - N b_q^G + b_q^S N.$$

The only possible additional codimension-one face consists of matching-line bubbles. We denote its signed contribution by B_q until Section 7.5. The desired integral boundary formula is therefore

$$(10) \quad N b_q^G - b_q^S N = d_S K_q^{p_1} + K_q^{p_1} d_G + B_q.$$

The remainder of the paper constructs this face and establishes $B_q = c_{g(q)} N$ with $c_1 = 8$, $c_2 = 4$, and $c_g = 0$ for $g \geq 3$.

3. INDEX-FOUR CONCENTRATION AND LOCAL CHARGE

3.1. Exhaustion of the index budget. Consider a sequence in $\mathcal{M}_\eta(\alpha, \beta)_r$. After DFL chain compactness, write (A_0, u_0) for the residual mixed solution, A_i^G for the nonconstant gauge Floer levels, and v_j^S for the nonconstant symplectic Floer levels. Define the global defect

$$(11) \quad \Delta = \mathcal{E}(A_\nu, u_\nu) - \mathcal{E}(A_0, u_0) - \sum_i \mathcal{E}(A_i^G) - \sum_j \mathcal{E}(v_j^S).$$

It includes all energy not carried by the displayed residual levels: ordinary bubbles, matching atoms, and any neck loss.

Proposition 3.1 (index-four exhaustion). *Assume that the residual moduli spaces are regular and have no negative-index solutions.*

- (1) *A sequence of mixed solutions of index at most three cannot bubble.*
- (2) *If an index-four sequence has $\Delta > 0$, then*

$$(12) \quad \Delta = \frac{1}{2}, \quad \text{ind } \mathcal{D}_{(A_0, u_0)} = 0,$$

and there are no nonconstant gauge or symplectic Floer levels.

(3) *The defect in part (2) carries all four missing Fredholm-index units.*

Proof. DFL compactness makes Δ nonnegative. If it is positive, the strict form of the topological-energy inequality and (5) give $\Delta \geq \frac{1}{2}$. Additivity of the mixed index under gluing gives

$$(13) \quad \text{ind } \mathcal{D}_{(A_0, u_0)} + \sum_i \text{ind } \mathcal{D}_{A_i^G} + \sum_j \text{ind } \mathcal{D}_{v_j^S} + 8\Delta \leq r.$$

Every term except the last is nonnegative, and each nonconstant Floer level has positive index. If $r \leq 3$, the inequalities $8\Delta \geq 4$ and (13) are incompatible. If $r = 4$, equality is forced throughout: $8\Delta = 4$ and every residual index other than the index-zero mixed component vanishes. A nonconstant Floer level cannot have index zero, proving the claim. $\square$

An interior four-dimensional instanton has integral charge at least one, hence energy at least one and index at least eight. It cannot occur in Proposition 3.1(2), either as the primary bubble or as a smaller-scale bubble inside a matching configuration. A nonconstant sphere in M_g has normalized energy at least $\frac{1}{2}$ and Chern index four; a nonconstant disc on either three-manifold Lagrangian has Maslov index at least four. Thus the half-unit budget permits at most one ordinary symplectic bubble and no simultaneous breaking.

3.2. Ordinary bubbles and the point incidence. The preceding numerical statement concerns the uncut moduli space. The point incidence removes the remaining ordinary sphere and disc strata from the finite- t boundary.

Proposition 3.2. *For a generic stratified representative of the universal p_1 cycle, no finite- t boundary point of $\bar{\mathcal{L}}_1^q(\alpha, \beta)$ contains an interior sphere bubble or an outer-boundary disc bubble.*

Proof. Put $2n = \dim_{\mathbb{R}} M_g$. The moduli space of simple minimal spheres with one nodal marked point has virtual dimension

$$(2n - 6) + 2c_1(A_g) + 2 = 2n,$$

because $c_1(A_g) = 2$. Constraining the nodal evaluation to the residual map removes $2n$ dimensions. The remaining two variables are the position of the node in the symplectic domain. Thus the stable-sphere stratum has codimension two in the uncut index-four mixed moduli space.

If the point incidence remains on the residual component, it is evaluated on the one-dimensional interpolation family of a fixed index-zero residual solution. A generic codimension-four cycle misses that family. If the incidence enters the sphere scale, requiring the bubble point to equal the point on the fixed interpolation path leaves one real domain variable; a marked point on the sphere adds two, and the p_1 incidence removes four. The resulting virtual dimension is $1 + 2 - 4 = -1$.

For an outer-boundary Maslov-four disc, the interpolation path is separated from the boundary, so the incidence cannot enter the disc scale. On the residual component it again sees only a one-dimensional family and is missed generically. Stable spheres or discs occurring after $t \rightarrow +\infty$ are part of the compactified symplectic point operation and are already included in $b_q^S N$; they are not finite- t correction terms. $\square$

The matching line is different. The interpolation path crosses the seam, so the ratio between its normal coordinate and the concentration scale is a parameter in the rescaled problem. Together with the center, scale, and marked-domain directions, it provides the four

real directions cut by the incidence. A matching bubble can therefore be a genuine boundary point of the one-dimensional cut.

3.3. The local Chern–Simons sector. DFL’s half-integrality controls the total defect, not each atom. The next theorem is the local quantization needed to rule out several matching centers whose a priori energy measures sum to $\frac{1}{2}$.

Let z be a matching point at which a sequence (A_ν, u_ν) concentrates. Choose a small half-disc on either side of the seam, with no other concentration point in its closure. On the negative semicircle use the restriction of A_ν . On the positive semicircle choose the flat representatives of u_ν supplied by the matching condition. At the two seam endpoints insert the standard flat cylinder. These pieces give a spliced connection

$$B_{\nu,r} \quad \text{on} \quad Y_r \cong S^1 \times \Sigma_g.$$

The removable limit gives a second connection $B_{0,r}$ on the same three-manifold.

DFL’s connection-valued filling construction replaces this boundary data by a smooth connection on $D^2 \times \Sigma_g$. Its Chern–Weil integral contains both the gauge energy and the symplectic area [DFL24, Lemma 4.67]; the equality between the curvature term of the flat-representative family and twice its symplectic area is [DFL26, Lemma 5.10].

Theorem 3.3 (local charge). *Let $\delta_z > 0$ be the atom of the normalized mixed energy measure at z . The original bundle and the removable limiting bundle determine a relative clutching integer n_z such that*

$$(14) \quad \delta_z = \frac{n_z}{2}, \quad n_z \in \mathbb{Z}_{>0}.$$

The sign is normalized so that the DFL positive generator has $n_z = 1$ and $p_1 = -2$.

Proof. Take regular radii $r \downarrow 0$ whose closed discs contain no other concentration point. For fixed r , compactness supplies annular gauges in which the boundary connections converge smoothly. Apply DFL’s filling construction to the original and removable limiting pairs, and denote the resulting bundles with connection over $D^2 \times \Sigma_g$ by $(Q_{\nu,r}, \mathbb{A}_{\nu,r})$ and $(Q_{0,r}, \mathbb{A}_{0,r})$. Use the limiting framing to identify $Q_{0,r}|_{Y_r}$ with a fixed bundle $Q_r^\partial \rightarrow Y_r$. The annular convergence gauge supplies a projective bundle isomorphism

$$\gamma_{\nu,r} : Q_{\nu,r}|_{Y_r} \longrightarrow Q_r^\partial$$

for which $\gamma_{\nu,r*}B_{\nu,r} \rightarrow B_{0,r}$ smoothly. No extension of $\gamma_{\nu,r}$ over either filling is chosen or required; a nonextendible gauge class will be the clutching class below. With DFL’s normalization, the filling identities just cited are

$$\mathcal{E}_\nu(r) = \frac{1}{8\pi^2} \int_{D^2 \times \Sigma_g} \text{tr}(F_{\mathbb{A}_{\nu,r}} \wedge F_{\mathbb{A}_{\nu,r}}), \quad \mathcal{E}_0(r) = \frac{1}{8\pi^2} \int_{D^2 \times \Sigma_g} \text{tr}(F_{\mathbb{A}_{0,r}} \wedge F_{\mathbb{A}_{0,r}}).$$

Interpolate linearly, in a fixed affine chart of connections on Q_r^∂ , between $\gamma_{\nu,r*}B_{\nu,r}$ and $B_{0,r}$ on $[0, 1] \times Y_r$. Smooth convergence makes its normalized Chern–Weil integral $\epsilon_{\nu,r}$ tend to zero. Glue the original filling through $\gamma_{\nu,r}$, this interpolation, and the reverse of the limiting filling. The result is an $SO(3)$ bundle $P_{\nu,r}$ over $S^2 \times \Sigma_g$. Thus every component of the annular gauge, including a possible large-gauge component, is recorded in $P_{\nu,r}$. Replacing $\gamma_{\nu,r}$ by a convergence gauge in the same local slice changes the clutching map by a homotopy and leaves $p_1(P_{\nu,r})$ unchanged. Chern–Weil excision, with the sign convention of [DFL26, Proposition 5.14], gives

$$(15) \quad \mathcal{E}_\nu(r) - \mathcal{E}_0(r) + \epsilon_{\nu,r} = -\frac{1}{4}p_1(P_{\nu,r}).$$

The manifold $S^2 \times \Sigma_g$ is spin. The Wu formula and $p_1(P) \equiv w_2(P)^2 \pmod{2}$ therefore show that $p_1(P_{\nu,r})$ is even. Put

$$n_{\nu,r} := -\frac{1}{2}p_1(P_{\nu,r}) \in \mathbb{Z}.$$

For fixed r , weak convergence of the energy measures and $\epsilon_{\nu,r} \rightarrow 0$ show that the half-integers $n_{\nu,r}/2$ converge. They are consequently eventually constant; write their limit as $n(r)/2$. As regular radii tend to zero,

$$\frac{n(r)}{2} = \lim_{\nu \rightarrow \infty} (\mathcal{E}_\nu(r) - \mathcal{E}_0(r)) \rightarrow \delta_z,$$

because the removable limit is smooth. The convergent half-integers $n(r)/2$ are again eventually constant. Calling the eventual integer n_z proves (14). Positivity of the atom gives $n_z > 0$. DFL's disc generator in Section 2.2 has normalized curvature integral $1/2$, so it has $p_1 = -2$ and fixes the stated sign.

All Chern–Simons lifts in this argument are the real-valued lifts selected by the displayed fillings. For two regular radii, restricting the original and limiting fillings to the intervening annulus gives the same clutching bundle on their common boundary. Hence changing the radius cannot transfer an integer to an independently chosen boundary branch; the only change is the Chern–Weil integral of that actual annular restriction, already present in (15). $\square$

The construction does not pass through M_g . It therefore retains the clutching class when $g = 1$, even though M_1 is a point. It is also additive over disconnected components of Σ , since the relative Pontryagin number splits componentwise.

Corollary 3.4 (unique primitive matching center). *Suppose an index-four sequence contributing to the point cut develops a matching atom. Then there is exactly one matching concentration point z , and*

$$(16) \quad n_z = 1, \quad \delta_z = \Delta = \frac{1}{2}.$$

There is no second matching point and no ordinary bubble or nonconstant Floer level away from z .

Proof. Proposition 3.1 gives total defect $\frac{1}{2}$. Theorem 3.3 assigns every matching atom a positive multiple of $\frac{1}{2}$. Thus the first atom already carries the whole defect and has charge one. Every ordinary sphere or disc also costs at least $\frac{1}{2}$, and every interior instanton costs at least one. Energy exhaustion excludes each such component away from the matching center. $\square$

Combining Proposition 3.2 and Corollary 3.4, the only new finite- t concentration locus is one charge-one matching center attached to an index-zero residual mixed solution. Quantization of the atom does not by itself exclude nested scales or neck energy at that same center. The next two sections prove the stable-reduction and gluing results needed to do so.

4. HOLOMORPHIC COMPACTIFICATION OF A PRIMITIVE ATOM

4.1. The local equations and base dilation. Choose coordinates (s, θ) centered at the matching point, with $D_- = \{s \leq 0\}$ the gauge half-disc and $D_+ = \{s \geq 0\}$ the symplectic half-disc. In a gauge adapted to the s direction, write the connection on $D_- \times \Sigma_g$ as

$$A = \beta(s, \theta) + \psi(s, \theta)d\theta.$$

The ASD equation is

$$(17) \quad *_\Sigma F_\beta + \partial_s \psi = 0,$$

$$(18) \quad -\partial_\theta \beta + *_\Sigma \partial_s \beta + d_\beta \psi = 0.$$

On D_+ the map u satisfies

$$(19) \quad \partial_\theta u - J_* \partial_s u = 0.$$

Along $s = 0$, the connection $\beta(0, \theta)$ is flat and represents $u(0, \theta)$.

For a concentration scale $\rho \rightarrow 0$, set

$$\beta_\rho(S, \Theta) = \beta(\rho S, \rho \Theta), \quad \psi_\rho(S, \Theta) = \rho \psi(\rho S, \rho \Theta).$$

Equations (17)–(18) become

$$(20) \quad \rho^2 *_\Sigma F_{\beta_\rho} + \partial_S \psi_\rho = 0,$$

$$(21) \quad -\partial_\Theta \beta_\rho + *_\Sigma \partial_S \beta_\rho + d_{\beta_\rho} \psi_\rho = 0.$$

The coefficient of vertical curvature is ρ^2 , not ρ^{-2} . This is the opposite of the Dostoglou–Salamon adiabatic regime and does not force the interior surface slices to be flat. The correct intermediate object is a holomorphic bundle on the product, rather than a direct map of the negative half-plane to M_g .

4.2. Finite-scale doubling. Give the base the complex coordinate $w = s + i\theta$.

Lemma 4.1 (finite-scale holomorphic doubling). *Every smooth mixed solution on a disc neighborhood of the matching line determines a holomorphic rank-two bundle on the full disc times Σ_g . It is canonical up to fixed-determinant complex gauge and tensoring by a line bundle pulled back from the disc.*

Proof. On the negative half, the product metric is Kähler. The ASD equation implies $F_A^{0,2} = 0$, so $A^{0,1}$ is an integrable Dolbeault operator.

Every seam fiber is stable. Stability is open, so the negative holomorphic family has stable fibers on a collar of the seam. The fine moduli property [Owe01, Section 4] gives a holomorphic classifying map f_- on that collar and identifies its bundle, up to a line bundle from the base, with the pullback of the universal bundle. The positive half similarly has the holomorphic classifying map $f_+ = u$.

The matching condition identifies the continuous boundary values of f_- and f_+ on the seam. In a complex chart of M_g , their union is continuous on a full collar and holomorphic on its two open halves. Morera’s theorem shows that it is holomorphic across the seam. Pull back the universal bundle by this glued map and use the resulting collar isomorphisms as transition maps between the negative and positive families. This gives the required holomorphic bundle. The fine-moduli ambiguity is a line bundle from the base; changing its trivialization and the collar isomorphisms has exactly the ambiguity stated in the lemma. $\square$

The point of Lemma 4.1 is that it is applied before taking the rescaled limit. No unproved trace convergence for a limiting connection-valued half-plane equation is required.

4.3. Stable reduction over a disc. The following one-parameter statement is the local stable-reduction input. It is the bundle-theoretic mechanism appearing in Owens’s cylindrical correspondence [Owe01].

Lemma 4.2 (disc stable reduction). *Let $E \rightarrow D \times \Sigma_g$ be a holomorphic rank-two bundle whose surface fibers have fixed determinant of degree one, and suppose at least one fiber is stable. Then:*

- (1) *the unstable fibers form a finite analytic subset of every compact subdisc;*
- (2) *the classifying map from the stable locus to M_g extends holomorphically across the unstable fibers without base change;*
- (3) *the original and pullback stable families differ by a finite Langton sequence supported on those fibers; all terms are locally free, retain the fixed surface-fiber determinant, and agree with E off those fibers;*
- (4) *if one step is the kernel of a quotient line G_j of degree $q_j \leq 0$, its positive charge is*

$$\ell_j = \frac{1}{2}(p_1(\text{ad } E^{j+1}) - p_1(\text{ad } E^j)) = 1 - 2q_j > 0.$$

Thus $\ell = \sum_j \ell_j$ is a nonnegative integer, and it vanishes exactly when no modification occurs.

Proof. The family E is flat over D . Toma's relative Harder–Narasimhan theorem makes semistability Zariski open in such an analytic family [Tom20, Theorem 2.4]. Here semistability is stability, so the existence of one stable fiber makes the unstable locus a proper analytic subset of D , hence finite on compact subdiscs.

Fix an unstable point and a smaller disc Δ containing no other one. Toma's analytic Langton theorem produces, without base change, a coherent subsheaf $E' \subset E$ that equals E over Δ^* , is flat over Δ , and has stable central fiber [Tom20, Theorem 3.1]. It is locally free. Indeed, all its fibers are torsion-free sheaves on the smooth curve and hence are vector bundles. At a stalk with base parameter t , lift a basis of E'/tE' to a map $R^2 \rightarrow E'$. Nakayama makes it surjective; flatness gives $\text{Tor}_1^R(E', R/(t)) = 0$, so Nakayama applied to the kernel makes the map an isomorphism.

In rank two Toma's construction is the finite Langton sequence

$$0 \longrightarrow E^{j+1} \longrightarrow E^j \longrightarrow i_* G_j \longrightarrow 0,$$

where, on the central curve, $B_j \subset E_0^j$ is the saturated destabilizing line and $G_j = E_0^j/B_j$. The transformed central fiber fits into

$$0 \longrightarrow G_j \longrightarrow E_0^{j+1} \longrightarrow B_j \longrightarrow 0.$$

Thus every step retains the determinant $B_j \otimes G_j$ and equals its predecessor off the central fiber. The final stable family supplies the holomorphic extension of the classifying map by the fine-moduli property; it differs from the pullback universal family only by a line bundle from Δ . In particular, the comparison preserves the outer attaching identification. This is also the local data in [Owe01, Proposition 5.2].

Put $q_j = \deg G_j$. Since $\deg E_0^j = 1$ and B_j destabilizes, $q_j \leq 0$. For the displayed elementary transform, the fiber divisor has square zero and

$$c_1(E^{j+1}) = c_1(E^j) - [\Sigma_g], \quad \text{ch}_2(E^{j+1}) = \text{ch}_2(E^j) - q_j.$$

It follows that

$$p_1(\text{ad } E^{j+1}) - p_1(\text{ad } E^j) = 2 - 4q_j.$$

This proves the charge formula and positivity. Repeating at the finitely many unstable fibers proves the lemma. $\square$

4.4. The primitive charge splitting. Let $E \rightarrow D \times \Sigma_g$ be one doubled family supplied by Lemma 4.1. The convergence to the removable residual solution fixes an outer projective framing. Apply Lemma 4.2, denote its final stable family by F , its classifying map by $f : D \rightarrow M_g$, and its total modification charge by ℓ .

Choose the outer circle so that $f(\partial D)$ lies in a contractible fine-moduli chart. Cap it in that chart and put

$$(22) \quad d = \left\langle -\frac{1}{2}p_1(\text{ad } \mathcal{U})/[\Sigma_g], [\widehat{f}(S^2)] \right\rangle.$$

This is the positive generator normalization: the pullback of $\text{ad } \mathcal{U}$ by a minimal line has Pontryagin number -2 [Muñ99b, Proposition 3].

Proposition 4.3 (fixed-family framed splitting). *If $n(E)$ is the relative clutching charge in the outer framing, then*

$$(23) \quad n(E) = d + \ell.$$

The identity is independent of the cap, the universal-bundle representative, the base-line ambiguity in the fine-moduli problem, and the factorization into elementary transforms. In the primitive sector, $n(E) = 1$, so the individual doubled family has either $(d, \ell) = (1, 0)$ or $(0, 1)$.

Proof. For one elementary transform with quotient line G of degree q , Grothendieck–Riemann–Roch and the triviality of the normal bundle of the fiber give

$$\text{ch}(i_*G) = [\Sigma_g] + q[\text{pt}].$$

Thus

$$c_1(E') = c_1(E) - [\Sigma_g], \quad \text{ch}_2(E') = \text{ch}_2(E) - q[\text{pt}].$$

Since

$$p_1(\text{ad } E) = 4 \text{ch}_2(E) - c_1(E)^2, \quad [\Sigma_g]^2 = 0, \quad c_1(E) \cdot [\Sigma_g] = 1,$$

one obtains

$$p_1(\text{ad } E') - p_1(\text{ad } E) = 2 - 4q = 2(1 - 2q).$$

All transforms are supported away from the framed outer circle, and hence

$$(24) \quad p_1(\text{ad } F) = p_1(\text{ad } E) + 2\ell$$

as a relative identity. The fine-moduli property identifies

$$F \cong (\text{id}_{\Sigma_g} \times f)^*\mathcal{U} \otimes p_D^*L_D.$$

The adjoint removes the base-line factor. After capping, (22) therefore says $p_1(\text{ad } \widehat{F}) = -2d$. By definition $p_1(\text{ad } \widehat{E}) = -2n(E)$, so (24) gives $n(E) = d + \ell$.

Two caps in the chosen chart are homotopic relative to the boundary. A base-line twist leaves the adjoint bundle unchanged, and $\ell = (p_1(\text{ad } F) - p_1(\text{ad } E))/2$ depends only on the framed endpoints. These observations prove all independence assertions. In the local primitive sector, Corollary 3.4 gives $n(E) = 1$. Nonnegativity and integrality of d and ℓ give the final dichotomy. $\square$

The compactness distinction. The pair (d, ℓ) belongs to one individual doubled family and need not be continuous under degeneration. A sequence with $(d, \ell) = (1, 0)$ can converge in the quasimap compactification to $(0, 1)$; the missing degree then appears as a sphere component in the stable-map compactification. That sphere and the quasimap basepoint are two records of the same primitive charge and cannot be added as independent components.

Near rank drop, write a full-rank pencil in an affine base coordinate as $s_\nu(z) = a_\nu z + b_\nu$. If its limiting basepoint is $z_\nu = x_\nu + iy_\nu$, choose a linear functional on the limiting image line, absorb the phase of its transverse coefficient into w_ν , and write exactly

$$(25) \quad s_\nu(z) = a_\nu(z - z_\nu) + \varepsilon_\nu w_\nu, \quad \varepsilon_\nu > 0, \quad \varepsilon_\nu \rightarrow 0, \quad a_\nu \rightarrow v, \quad w_\nu \rightarrow w, \quad v \wedge w \neq 0.$$

After the seam-preserving change $z = \varepsilon_\nu \zeta + iy_\nu$, the pencil is

$$[a_\nu(\zeta - \lambda_\nu) + w_\nu], \quad \lambda_\nu = x_\nu/\varepsilon_\nu.$$

If λ_ν has a finite limit, this converges to a full-rank mixed bubble. If $\lambda_\nu \rightarrow +\infty$ or $-\infty$, it is constant on seam-centered compact sets and recentering at z_ν extracts, respectively, a positive holomorphic sphere or a negative cylindrical instanton. Thus a seam basepoint is not a fourth type: center-scale normalization either recovers a mixed bubble or one of the two pure ends.

The required analytic statement is strong convergence of arbitrary primitive mixed sequences, in seam-compatible gauges and the first-order norms used below, to this three-colored master compactification. At negative infinite seam ratio, affine recentering returns to the normalized gauge rank-drop chart. At positive infinite ratio the additional component is an ordinary minimal sphere, whose nodal collar must be matched to the finite-ratio charts. Sections 6.4–6.6 prove this analytic realization using good-circle no-neck, the uniform anisotropic inverse, and local surjectivity of the gluing maps.

5. THE ALGEBRAIC PRIMITIVE MODULI

5.1. Muñoz’s extension spaces. Fix a line bundle Λ of degree one on Σ_g . Let $J = J(\Sigma_g)$ be the Jacobian and let $\mathcal{L}$ be a Poincaré line bundle on $\Sigma_g \times J$. If $p : \Sigma_g \times J \rightarrow J$ is the projection, put

$$(26) \quad \mathcal{E}_g = R^1 p_*(\mathcal{L}^2 \otimes \Lambda^{-1}).$$

With the standard principal polarization $\omega \in H^2(J; \mathbb{Z})$, the Grothendieck–Riemann–Roch calculation is

$$(27) \quad \text{ch}(\mathcal{E}_g) = g + 4\omega.$$

The projective bundle

$$N_g = \mathbb{P}(\mathcal{E}_g^\vee) \longrightarrow J$$

parametrizes nonsplit extensions

$$(28) \quad 0 \longrightarrow L \longrightarrow V \longrightarrow \Lambda \otimes L^{-1} \longrightarrow 0.$$

The middle term is stable and determines a holomorphic map

$$(29) \quad i_g : N_g \longrightarrow M_g.$$

Muñoz studies stable rank-two bundles E on $\mathbb{P}^1 \times \Sigma_g$ with

$$(30) \quad c_1(E) = [\mathbb{P}^1] + [\Sigma_g], \quad c_2(E) = 1, \quad p_1(\text{ad } E) = -2,$$

for a polarization close to the Σ_g boundary of the ample cone. He proves that their moduli space is the smooth projective bundle

$$(31) \quad \mathfrak{M}_g = \mathbb{P}((\mathcal{E}_g \oplus \mathcal{E}_g)^\vee) \longrightarrow J, \quad \dim_{\mathbb{C}} \mathfrak{M}_g = 3g - 1.$$

See [Muñ99b, Sections 3–4]. The two summands are the coefficients of a section of $\mathcal{O}_{\mathbb{P}^1}(1) \otimes \mathcal{E}_g$.

Fix $L \in J$, write $V = (\mathcal{E}_g)_L$, and represent a point of the fiber by

$$[v_0 : v_1] \in \mathbb{P}(V \oplus V).$$

At $x = [x_0 : x_1] \in \mathbb{P}^1$, the corresponding surface extension has class

$$(32) \quad v(x) = x_0 v_0 + x_1 v_1.$$

Proposition 5.1 (rank stratification). *The moduli space $\mathfrak{M}_g$ has the following two strata.*

- (1) *If v_0, v_1 are independent, (32) never vanishes and defines a nonconstant parameterized minimal rational curve in M_g . Every parameterized minimal rational curve arises uniquely in this way.*
- (2) *If v_0, v_1 are dependent, write $(v_0, v_1) = (a_0 v, a_1 v)$. The induced quasimap is constant away from the single basepoint $[-a_1 : a_0]$. The rank-drop locus is the Segre bundle*

$$(33) \quad \mathcal{R}_g = N_g \times_J \mathbb{P}^1 \longrightarrow \mathfrak{M}_g$$

of complex codimension $g - 1$.

For an individual primitive bundle, the two alternatives in Proposition 4.3 correspond respectively to these two strata.

Proof. If the two vectors are independent, no nonzero pair (x_0, x_1) kills their linear combination. Thus (32) defines a line in the projective extension fiber $\mathbb{P}(V)$, followed by the map i_g . Muñoz’s classification identifies this full-rank open locus with the space of parameterized minimal curves.

If the two vectors are dependent, every nonzero extension away from $[-a_1 : a_0]$ represents the same point $[v] \in \mathbb{P}(V)$. At the unique zero the surface bundle splits, and stable reduction records one elementary modification. Fiberwise, the dependent tensors form the Segre image $\mathbb{P}(V) \times \mathbb{P}(\mathbb{C}^2)$ in $\mathbb{P}(V \otimes \mathbb{C}^2)$. Its dimension is $(g - 1) + 1 = g$, whereas the ambient projective space has dimension $2g - 1$, so its codimension is $g - 1$. The final assertion follows from the exact dichotomy in Proposition 4.3. $\square$

For $g = 1$, the extension space V is one-dimensional, and every pair is dependent. The full-rank locus is empty and $\mathfrak{M}_1 = \mathcal{R}_1$. This is the algebraic reason the genus-one primitive class is invisible in the point-valued character variety.

5.2. Evaluation at infinity and the dimension ledger. Choose $\infty = [1 : 0] \in \mathbb{P}^1$. On the locus $v_0 \neq 0$, evaluation of the induced stable map or quasimap is

$$(34) \quad \text{ev}_\infty([v_0 : v_1]) = i_g([v_0]).$$

Hence every primitive bubble, full rank or rank drop away from infinity, attaches through $i_g(N_g)$.

Fix $[v] \in N_g$. After the overall projective scalar is used to set $v_0 = v$, the full-rank fiber of ev_∞ is

$$V \setminus \mathbb{C}v.$$

The real affine group preserving the two half-planes and fixing infinity is

$$(35) \quad G_{\mathbb{R}} = \{w \longmapsto aw + ib : a > 0, b \in \mathbb{R}\}.$$

It fixes the translation and scale of a generic mixed bubble. The reduced fiber therefore has real dimension $2g - 2$.

The dimensions of the attachment spaces are

$$(36) \quad \dim_{\mathbb{C}} N_g = 2g - 1, \quad \dim_{\mathbb{C}} M_g = 3g - 3.$$

Muñoz proves that i_g is generically finite onto its image. Thus

$$(37) \quad \text{codim}_{\mathbb{R}} i_g(N_g) = 2(g - 2).$$

Let $e_0 : \mathbb{R} \rightarrow M_g$ be the matching evaluation path of a residual index-zero mixed solution. Whenever it meets the regular part of $i_g(N_g)$, the reduced full-rank primitive stratum has dimension

$$(38) \quad (2g - 2) + 1 - 2(g - 2) = 3.$$

The interpolation parameter adds one dimension and the integral p_1 incidence removes four, leaving isolated boundary points.

For $g \geq 3$, however, a generic one-dimensional path misses the positive-codimension image in (37). The matching correction is therefore absent. In genus two, N_2 and M_2 have the same dimension and i_2 has a nonzero degree, computed next.

5.3. The genus-two degree. Let $h = c_1(\mathcal{O}_{N_2}(1))$, and normalize the principal polarization by

$$(39) \quad \int_{J(\Sigma_2)} \frac{\omega^2}{2} = 1.$$

If $\alpha \in H^2(M_2; \mathbb{Z})$ is the standard generator, Muñoz's universal-bundle formulas give

$$(40) \quad i_2^* \alpha = 4\omega + h, \quad h^2 + 4\omega h + 8\omega^2 = 0.$$

Proposition 5.2. *The map $i_2 : N_2 \rightarrow M_2$ has degree four.*

Proof. Using (40) and $\omega^3 = 0$ gives

$$h^3 = 8\omega^2 h, \quad 12\omega h^2 = -48\omega^2 h.$$

Therefore

$$(41) \quad \int_{N_2} (4\omega + h)^3 = \int_{N_2} 8\omega^2 h = 16,$$

where the last equality uses (39) and integration over the projective-line fiber. The genus-two moduli space M_2 is a smooth intersection of two quadrics in $\mathbb{P}^5$, and α is its hyperplane class. Hence

$$\int_{M_2} \alpha^3 = 4.$$

The ratio of (41) to this number is the degree: $\deg i_2 = 16/4 = 4$. □

All local algebraic multiplicities are positive in the complex orientation. Thus, after the analytic incidence determinant is identified with its algebraic counterpart, the magnitude of the integral genus-two contribution is four.

5.4. **The genus-one excess line.** For $g = 1$, (27) reads

$$(42) \quad \text{ch}(\mathcal{E}_1) = 1 + 4\omega.$$

The two coefficients of the primitive extension form the rank-two bundle

$$(43) \quad \mathcal{E}_\zeta = H^0(\mathbb{P}^1, \mathcal{O}(1)) \otimes \mathcal{E}_1 \cong \mathcal{E}_1 \oplus \mathcal{E}_1.$$

Consequently

$$(44) \quad c_1(\det \mathcal{E}_\zeta) = 8\omega, \quad \int_{J(\Sigma_1)} c_1(\det \mathcal{E}_\zeta) = 8.$$

After the rank-drop basepoint and the real affine symmetry are fixed, the elliptic Jacobian is the excess primitive family. Section 7.4 identifies its normal incidence determinant with $\det \mathcal{E}_\zeta$. Thus (44) is the magnitude of the integral genus-one contribution.

5.5. **Cylinder calibration.** The algebraic magnitudes can be calibrated, including sign, without assuming the point theorem. Let a_G, b_G denote Muñoz’s surface and point generators in the instanton Floer ring of $\Sigma_g \times S^1$, and let a_S, b_S be the geometric generators of $QH^*(M_g)$. These act on the gauge and symplectic Floer groups of every admissible target containing the chosen surface component.

We record the normalization dictionary. Muñoz uses $\mu(q) = -\frac{1}{4}p_1/[q]$ and defines his point generator by $b = -4\mu(q) = p_1/[q]$; his surface generator is $a = 2\mu(\Sigma) = -\frac{1}{2}p_1/[\Sigma]$ [Muñ99a, equation (4)]. Thus b_G, b_S are exactly the positive integral point generators used here. DFL’s comparison for $p_1/[\Sigma] = -2a$ implies, over $\mathbb{Q}$,

$$(45) \quad Na_G = a_S N.$$

For genus two, Muñoz’s corrected quantum generator is $\widehat{b}_S = b_S + 4$, and his relation $a_S^2 + \widehat{b}_S - 8 = 0$ is therefore $a_S^2 + b_S - 4 = 0$ in the unshifted geometric generator [Muñ99b, Example 24]. On the gauge side his corresponding relation is $a_G^2 + b_G - 8 = 0$ [Muñ99a, Example 22]. In genus one his gauge relation is $b_G = 8 \text{ id}$ [Muñ99a, Lemma 11], whereas the geometric class b_S vanishes because M_1 is a point [Muñ99b, Example 23].

Proposition 5.3 (low-genus cylinder identities). *In the positive integral p_1 normalization,*

$$(46) \quad Nb_G - b_S N = \begin{cases} 8N, & g = 1, \\ 4N, & g = 2. \end{cases}$$

Proof. For $g = 1$, the odd-determinant character variety is a point, so $b_S = 0$. Muñoz’s cylinder presentation has $b_G = 8 \text{ id}$, proving the first formula.

For $g = 2$, the instanton and quantum relations in the positive integral normalization are

$$a_G^2 + b_G - 8 = 0, \quad a_S^2 + b_S - 4 = 0.$$

Acting on the two modules and using (45) twice gives

$$\begin{aligned} Nb_G - b_S N &= N(8 - a_G^2) - (4 - a_S^2)N \\ &= 4N - a_S^2 N + Na_G^2 = 4N. \end{aligned}$$

□

The proposition already proves that uncorrected point compatibility is false in low genus. In the final orientation argument it has a second, noncircular role: determinant excision makes the local bubble coefficient universal, and (46) fixes the universal DFL sign.

6. DIRICHLET RECONSTRUCTION AND PRIMITIVE GLUING

6.1. Seam-admissible product bundles. Decompose the base sphere into closed hemispheres

$$(47) \quad \mathbb{P}^1 = H_- \cup_{\Gamma} H_+,$$

where Γ is the compactified matching line and $\infty \in \Gamma$ is the attaching point. The gauge equation is placed on $H_- \times \Sigma_g$ and the holomorphic-map equation on H_+ . A primitive bundle $E \in \mathfrak{M}_g$ induces on the stable fibers of the base a classifying map to M_g .

Definition 6.1. A bundle $E \in \mathfrak{M}_g$ is *seam admissible* if every restriction $E|_{\{z\} \times \Sigma_g}$ with $z \in \Gamma$ is stable and the classifying quasimap has no basepoint in H_+ . We denote the seam-admissible locus by $\mathfrak{M}_g^{\text{sa}}$.

Every full-rank pair is seam admissible. A rank-drop pair is seam admissible precisely when its unique basepoint lies in the interior of H_- . A basepoint in H_+ makes the positive map undefined, while a basepoint on Γ cannot occur in a smooth seam-admissible object. For a sequence approaching Γ , the center-scale normal form (25) either recovers a full-rank mixed bubble at finite seam ratio or converges to one of the two pure limits. The real Möbius group preserving (47) acts transitively on the interior of H_- ; after fixing infinity, its affine subgroup $G_{\mathbb{R}}$ in (35) normalizes a gauge-side basepoint to a compact set.

6.2. Exact reconstruction. For $\rho > 0$, equip the gauge hemisphere with the product Kähler form

$$(48) \quad \omega_{\rho} = \rho^2 \omega_{H_-} + \omega_{\Sigma_g}.$$

This is the form obtained by pulling the original metric back under base dilation by ρ .

Theorem 6.2 (Dirichlet reconstruction). *Let $E \in \mathfrak{M}_g^{\text{sa}}$. For every $\rho > 0$ there is a unique fixed-determinant projectively ASD connection $A_{E,\rho}$ on $E|_{H_- \times \Sigma_g}$ whose boundary metric is the family of Narasimhan–Seshadri metrics on the stable seam fibers. Together with the holomorphic classifying map*

$$u_E : H_+ \longrightarrow M_g,$$

it is an exact solution of the ρ -rescaled mixed equation. It has relative clutching charge one and normalized mixed energy $\frac{1}{2}$. For $\rho > 0$ the solution depends smoothly on (E, ρ) .

Proof. Each stable odd-degree bundle on a seam fiber has a unique fixed-determinant Hermitian–Einstein metric. Simplicity removes every noncentral ambiguity, and elliptic parameter dependence makes these metrics a smooth family H_{Γ} on $E|_{\Gamma \times \Sigma_g}$.

Donaldson’s boundary-value theorem [Don92, Theorem 1] gives a unique Hermitian metric $H_{E,\rho}$ with fixed determinant satisfying

$$(49) \quad H_{E,\rho}|_{\Gamma \times \Sigma_g} = H_{\Gamma}, \quad i\Lambda_{\omega_{\rho}} F_{H_{E,\rho}}^{\perp} = 0.$$

On a Kähler surface, integrability and the trace-free Hermitian Yang–Mills equation are equivalent to projective anti-self-duality. The Chern connection of $H_{E,\rho}$ is therefore the required $A_{E,\rho}$. No stability of E with respect to ω_{ρ} is needed: the Dirichlet theorem applies to every holomorphic bundle on a compact Kähler manifold with boundary. This point is essential because the polarization (48) approaches the chamber opposite to the one used to define $\mathfrak{M}_g$.

On H_+ , seam admissibility gives the holomorphic map u_E . Its pullback universal bundle is $E|_{H_+ \times \Sigma_g}$, and its canonical flat representative along Γ is the Chern connection of H_{Γ} .

Thus the two sides satisfy the exact matching condition. Holomorphicity gives the positive equation.

The Chern–Weil filling identity and $p_1(\text{ad } E) = -2$ give relative charge one and energy $\frac{1}{2}$. The linearization of (49) in the Hermitian metric direction is the Dirichlet Chern Laplacian on trace-free self-adjoint endomorphisms. Its kernel is zero by integration by parts. Elliptic parameter dependence and the implicit-function theorem give smooth dependence on (E, ρ) . $\square$

For a full-rank E , the positive map is part of a basepoint-free minimal line and the negative half is its unique Dirichlet ASD completion. For a rank-drop E with basepoint in H_- , the positive map is constant and the entire modification lies on the gauge hemisphere. Thus every individual algebraic type in Proposition 4.3 is realized by an exact local mixed solution. In genus one the second type is the pure gauge bubble.

6.3. Fredholm reduction. Fix $E \in \mathfrak{M}_g^{\text{sa}}$ and $\rho > 0$. Infinitesimal deformations of the reconstructed pair split into a Dolbeault deformation of E and a trace-free Hermitian-metric deformation with zero Dirichlet value. The metric block is invertible by the last paragraph of the proof of Theorem 6.2. Domain reparameterizations are added only after this fixed-domain problem is coupled to the residual solution.

Proposition 6.3 (fixed-scale Fredholm-to-Dolbeault reduction). *Let $\mathcal{D}_{E,\rho}^{\text{mix}}$ be DFL’s gauge-fixed mixed operator, with the relative boundary condition and canonical linearized matching correspondence in its domain. Infinitesimal holomorphic doubling gives a natural real-linear isomorphism*

$$(50) \quad \ker \mathcal{D}_{E,\rho}^{\text{mix}} \cong H^1(\mathbb{P}^1 \times \Sigma_g, \text{End}_0 E).$$

The mixed operator is surjective. Hence the fixed-domain mixed bubble is regular and its determinant line has the complex orientation transported through (50). Evaluation at infinity and quotienting by $G_{\mathbb{R}}$ are separate finite-dimensional operations imposed only after coupling to the residual component.

Proof. DFL’s operator has components

$$(d_A^+ \zeta, -d_A^* \zeta, D_u \nu).$$

Its domain satisfies $*\zeta|_{\Gamma} = 0$ and $(\zeta|_{\{z\} \times \Sigma_g}, \nu(z)) \in T\mathcal{L}_{\text{NS}}$, while its formal adjoint uses the same canonical correspondence [DFL24, (1.11), (1.13), and (5.1)]. Their local estimates, patched over the compact hemispheres, make this a Fredholm boundary problem.

DFL regularity makes every kernel element smooth. The Kähler identity identifies the negative component $a^{0,1}$ with a Dolbeault cocycle. On the positive side, harmonic inclusion for the universal connection identifies a map variation with the infinitesimal deformation of the pulled-back universal bundle. The matching condition says that the two seam classes agree; their difference is $\bar{\partial}_\alpha \xi$ for a smooth infinitesimal complex gauge transformation. Extend ξ smoothly across a collar and differentiate the finite-scale doubling construction of Lemma 4.1. This produces the class in the right side of (50). Different extensions change it by a Dolbeault coboundary.

If this class vanishes, the positive harmonic variation is zero and the negative variation changes only the Hermitian metric of a fixed holomorphic bundle. Its boundary value is zero. The linearized Dirichlet HYM equation has

$$\langle L_{E,\rho}^{\text{Dir}} s, s \rangle = 2 \|\bar{\partial}_{A_{E,\rho}} s\|^2.$$

It has zero kernel, proving injectivity. Conversely, Muñoz identifies the primitive moduli with the smooth projective bundle $\mathbb{P}(\mathcal{E}_\zeta^\vee) \rightarrow J(\Sigma_g)$. Integrate a class in H^1 to a curve in this projective bundle, apply Theorem 6.2 at the fixed scale, and differentiate. This proves surjectivity in (50).

It remains to rule out a DFL cokernel. Deform the coefficients to product form in a seam collar and use DFL's bounded domain-straightening maps [DFL24, Proposition 5.27]. The Kähler identities identify the gauge symbol and Green form with the Dolbeault–Dirac symbol and Green form. Fiberwise Hodge decomposition pairs every nonzero exact vertical mode with its coexact mate; the two half-line operators are adjoints and have cancelling indices. Irreducibility removes the zero exact mode. The remaining harmonic matching condition is the graph transmission condition. Its Sobolev domain is the domain of the uncut closed operator [BB16, Example 3.23]. Relative index gluing therefore gives

$$\mathrm{ind}_{\mathbb{R}} \mathcal{D}_{E,\rho}^{\mathrm{mix}} = \mathrm{ind}_{\mathbb{R}} \sqrt{2}(\bar{\partial}_E^* \oplus \bar{\partial}_E) = 2(h^1 - h^0 - h^2).$$

Stability gives $h^0 = 0$, while Riemann–Roch gives

$$h^1 - h^2 = 3g - 1.$$

The smooth projective-bundle description gives $h^1 = 3g - 1$, including on the Segre locus. Thus the real kernel dimension equals the real index, and the cokernel is zero. DFL's adjoint regularity and its identical Lagrangian matching condition show that this is the analytic cokernel. Since the cokernel vanishes, (50) transports the complex orientation to the determinant line. $\square$

The rank-drop normal can be read explicitly. At a tensor $[v \otimes a] \in \mathbb{P}(V \otimes \mathbb{C}^2)$, let $\ell_v = \mathbb{C}v$ and $\ell_a = \mathbb{C}a$. The normal to the Segre locus is

$$(51) \quad \mathrm{Hom}(\ell_v, V/\ell_v) \otimes \mathrm{Hom}(\ell_a, \mathbb{C}^2/\ell_a),$$

of complex dimension $g - 1$. In genus two it is a complex line; after its basepoint is fixed, both real components remain as smoothing directions from rank drop to a full-rank bubble. The seam-preserving affine group has no rotational factor and does not remove the phase. Thus the genus-two Segre locus has real codimension two inside the primitive face. In genus one the normal has rank zero, leaving the elliptic excess family.

6.4. Primitive no-neck at good circles. Let (A_ν, u_ν) be a sequence with one primitive matching center $z_\nu \rightarrow z_0$ and concentration scales $\rho_\nu \rightarrow 0$, and suppose its doubled bundles have a limit E in the three-colored master compactification described after Proposition 4.3. Write $\mathcal{E}_\nu(a, b)$ for the normalized mixed energy in the paired half-annulus $a < |z - z_\nu| < b$.

Proposition 6.4 (primitive good-circle no-neck). *After passage to a subsequence, there are radii*

$$\rho_\nu \ll a_\nu \ll b_\nu \longrightarrow 0$$

such that

$$(52) \quad \mathcal{E}_\nu(a_\nu, b_\nu) \longrightarrow 0.$$

The capped region inside a_ν has clutching sector one and energy tending to $\frac{1}{2}$. Hence no positive-energy atom or further analytic scale lies between the primitive core and the residual solution. On the two cut circles, the spliced representatives converge in a common seam gauge to the same flat attaching representative.

Proof. The stable limit is regular at infinity. For a sufficiently large rescaled radius R , cap its tail in a contractible fine-moduli chart. The finite charge identity in Proposition 4.3 shows that, for R and then ν large, the capped region $|z - z_\nu| \leq R\rho_\nu$ has clutching sector one. This includes rank drop because the lost map degree is exactly the length-one elementary modification. The local charge theorem gives the same sector for the outer atom. Every annulus between the two caps therefore has sector zero.

Put $E_\nu(r) = \mathcal{E}_\nu(0, r)$. First choose the outer cuts from smooth convergence on compact subsets of the punctured disc to the residual solution. DFL removability extends that solution over z_0 , so a diagonal choice gives $b_\nu \rightarrow 0$, with $b_\nu/\rho_\nu \rightarrow \infty$, such that the outer-tail energy relative to the residual solution and $b_\nu E'_\nu(b_\nu)$ both tend to zero.

Since

$$\int_{r_1}^{r_2} r E'_\nu(r) d \log r = E_\nu(r_2) - E_\nu(r_1),$$

choose $R_\nu \rightarrow \infty$ slowly enough that $R_\nu^2 \rho_\nu < b_\nu$, and subdivide the logarithmic region from $R_\nu \rho_\nu$ to $R_\nu^2 \rho_\nu$ into dyadic shells. Bounded total energy and coarea give a shell of energy tending to zero and a radius a_ν in it for which

$$(53) \quad a_\nu E'_\nu(a_\nu) + b_\nu E'_\nu(b_\nu) \rightarrow 0.$$

At either good circle form DFL's spliced connection $B_{\nu,r}$ on $S^1 \times \Sigma_g$: it joins the gauge semicircle to a connection representing the holomorphic semicircle through the flat interpolation strip. The small-energy mean-value estimate on the selected shell and the proof of [DFL24, Lemma 4.62] give the localized slice bound

$$|\text{CS}_{\beta_{\nu,r}}(B_{\nu,r})| \leq Cr E'_\nu(r).$$

The flat references at the two radii are joined by the seam path, so the Chern–Simons functional is independent of that choice. The Stokes calculation of [DFL24, Lemma 4.67], now on the annulus, gives

$$\mathcal{E}_\nu(a_\nu, b_\nu) = \text{CS}(B_{\nu,b_\nu}) - \text{CS}(B_{\nu,a_\nu})$$

with the normalization of (3). The possible integral change of lift is zero because the annulus has clutching sector zero. Equation (53) proves (52).

The total local energy tends to $\frac{1}{2}$, the annular energy vanishes, and the outer-tail energy tends to that of the removable residual solution. Thus the sector-one inner core has energy $\frac{1}{2}$ in the limit. A further atom would have positive half-integral charge, whereas a diffuse remainder is excluded by the same Chern–Simons identity. Finally, DFL's small-energy Coulomb gauges on the two selected shells converge to the flat seam representatives; stable reduction at infinity identifies both with the attaching value. $\square$

6.5. Inverse-adiabatic Dirichlet estimates. Put $\tau = \rho^2$. Since the Kähler form is a product, the trace-free HYM equation is equivalent to

$$(54) \quad i\Lambda_{H_-} F_H^\perp + \tau i\Lambda_{\Sigma_g} F_H^\perp = 0.$$

Thus the limit $\tau = 0$ is elliptic in the base variable and regards the surface variable as a parameter. This is the useful direction of the inverse-adiabatic limit.

For the following statements, $W_{B,\Sigma}^{\tau,s;p}$ controls mixed derivatives with at most r base derivatives and at most s surface derivatives in fixed product charts. On the HYM-equation component we use the equivalent ρ -weighted target norm obtained by multiplying the equation by ρ^2 , as in (54). Fixed weights are added at the attaching end.

Theorem 6.5 (inverse-adiabatic Dirichlet regularity). *Let $C \subset \mathfrak{M}_g^{\text{sa}}/G_{\mathbb{R}}$ be compact. The Dirichlet metrics $H_{E,\rho}$ extend smoothly in (E, τ) to $\tau = 0$ in every anisotropic $W_{B,\Sigma}^{r,s;p}$ space. The limiting metric $H_{E,0}$ is the unique fixed-determinant metric satisfying*

$$(55) \quad i\Lambda_{H_-} F_{H_{E,0}}^\perp = 0, \quad H_{E,0}|_{\Gamma \times \Sigma_g} = H_\Gamma.$$

In particular, relative to a smooth reference family, the metrics, connections, and all fixed-coordinate derivatives are bounded uniformly for $E \in C$ and $0 \leq \rho \leq \rho_0$. This conclusion holds through the rank-drop locus.

Proof. Fix $x \in \Sigma_g$. The restriction of E to $H_- \times \{x\}$ is a holomorphic bundle on a disc. Donaldson's Dirichlet theorem in complex dimension one gives the unique fixed-determinant solution of (55) with the prescribed boundary metric. Equivalently, it is the harmonic metric for a disc with target the nonpositively curved symmetric space of positive Hermitian matrices of determinant one. Ordinary parameter-dependent Dirichlet theory makes these solutions smooth in (E, x) and gives a smooth metric $H_{E,0}$ on the product.

Write nearby metrics as $H_{E,0}e^s$, where s is trace-free, self-adjoint, and zero on the boundary. The derivative at $(s, \tau) = (0, 0)$ of the left side of (54) is the base Dirichlet Chern Laplacian

$$L_{E,0}^B s = i\Lambda_{H_-} \bar{\partial}_B \partial_{H_{E,0}, B} s.$$

For each x , integration by parts identifies its quadratic form with $2\|\bar{\partial}_B s\|^2$. Its kernel is zero by the boundary condition, and the Dirichlet operator on the disc has index zero. It is therefore an isomorphism. Differentiating in the parameter x gives the standard parameter-elliptic estimates, so

$$L_{E,0}^B : W_{B,\Sigma,0}^{r+2,s;p} \longrightarrow W_{B,\Sigma}^{r,s;p}$$

is an isomorphism for every r, s and $p > 1$.

It remains to justify uniformity for the exact solutions rather than use an implicit-function theorem with a surface-derivative loss. Put $h_\rho = H_{E,0}^{-1} H_{E,\rho}$. Because $H_{E,0}$ is base-flat, $\Lambda_{\omega_\rho} F_{H_{E,0}}^\perp$ is bounded independently of ρ . The Chern-connection trace inequality gives

$$-\Delta_{\omega_\rho} \log \text{tr } h_\rho \leq C.$$

If $-\Delta_{H_-} \phi = 1$ and $\phi|_\Gamma = 0$, then $-\Delta_{\omega_\rho}(\rho^2 \phi) = 1$. Since $h_\rho = \text{id}$ on the boundary, the maximum principle gives

$$(56) \quad \log \text{tr } h_\rho \leq \log 2 + C\rho^2.$$

The determinants agree, so (56) makes $s_\rho = \log h_\rho$ uniformly small in C^0 .

This initial smallness gives the sharper order in τ . Define

$$\Phi(H, \tau) = i\Lambda_{H_-} F_H^\perp + \tau i\Lambda_{\Sigma_g} F_H^\perp.$$

The fundamental theorem of calculus along $H_{E,0}e^{ts_\rho}$ gives the exact equation

$$(57) \quad \left(\int_0^1 D_H \Phi(H_{E,0}e^{ts_\rho}, \tau) dt \right) s_\rho = -\tau i\Lambda_{\Sigma_g} F_{H_{E,0}}^\perp.$$

For ρ small, the averaged operator has the same uniform base Dirichlet coercivity as $L_{E,0}^B$; the C^0 smallness permits its coefficient error to be absorbed. Base Dirichlet regularity therefore first gives $\|s_\rho\|_{W_{B,\Sigma}^{2,0;p}} \leq C\tau$.

Apply surface and parameter difference quotients to (57). At every order the highest derivative v satisfies the linearized Dirichlet equation, whose normalized quadratic form obeys

$$(58) \quad \|\nabla_B v\|_{L^2}^2 + \tau \|\nabla_\Sigma v\|_{L^2}^2 \leq C(\|f\|_{H_B^{-1}}^2 + \|v\|_{L^2}^2).$$

Here f contains only lower-order difference quotients. The base Dirichlet Poincaré inequality absorbs the last term, with a constant independent of τ . Difference-quotient induction, followed by base Dirichlet regularity and Sobolev embedding, gives

$$\|s_\rho\|_{W_{B,\Sigma}^{r,s;p}} \leq C_{r,s,p} \tau$$

at every finite pair of orders. At each induction step, derivatives of the averaged operator multiply already controlled lower derivatives of s_ρ , while the differentiated right side of (57) retains its factor τ . Differentiating the equation in τ and repeating the same estimate gives all τ derivatives. Compactness of C makes all constants uniform. This proves the claimed smooth extension, and Donaldson uniqueness identifies it with the family in Theorem 6.2. Rank drop creates an unstable *surface* fiber, but the limiting Dirichlet operator acts in the base disc, where every restricted bundle is holomorphically trivial; hence it introduces no exceptional model. $\square$

Proposition 6.6 (uniform metric-block coercivity). *Let $L_{E,\rho}^{\text{Dir}}$ be the trace-free self-adjoint Dirichlet Chern Laplacian at a reconstructed bubble. There is a constant $c > 0$, independent of $E \in \mathfrak{M}_g^{\text{sa}}$ and $0 < \rho \leq 1$, such that*

$$(59) \quad \langle L_{E,\rho}^{\text{Dir}} s, s \rangle \geq c\rho^{-2} \|s\|_{L^2(\omega_\rho)}^2, \quad s|_{\Gamma \times \Sigma_g} = 0.$$

Consequently its L^2 inverse has norm at most $c^{-1}\rho^2$.

Proof. The Bochner–Kodaira identity on endomorphisms gives

$$2\|\bar{\partial}_{A_{E,\rho}} s\|^2 = \|\nabla_{A_{E,\rho}} s\|^2 + \langle i[\Lambda_{\omega_\rho} F_{A_{E,\rho}}, s], s \rangle.$$

The last term vanishes because the connection is projectively HYM. Kato’s inequality and the scalar Dirichlet Poincaré inequality on H_- imply

$$\|\nabla_{A_{E,\rho}} s\|_{L^2(\omega_\rho)}^2 \geq \|d|s|\|_{L^2(\omega_\rho)}^2 \geq c\rho^{-2} \|s\|_{L^2(\omega_\rho)}^2.$$

This argument contains no curvature bound and is therefore unaffected by rank drop. $\square$

Theorem 6.7 (uniform reduced inverse). *Let C be a compact subset of $\mathfrak{M}_g^{\text{sa}}/G_{\mathbb{R}}$. On the complement of the Dolbeault tangent space, the mixed linearization at the reconstructed bubbles has a right inverse $Q_{E,\rho}$ satisfying*

$$(60) \quad \|Q_{E,\rho}\| \leq C_0$$

for $E \in C$ and $0 < \rho \leq \rho_0$.

Proof. Fix $p > 4$ and put $\tau = \rho^2$. We first specify the norms in the statement. After the collar Cauchy–Riemann identification below, write s for the full coexact–metric pair and χ for the full exact–unitary–gauge pair; each displayed scalar norm is applied componentwise. The first pair, with zero Dirichlet value, has norm

$$\begin{aligned} \|s\|_{\mathcal{M}_\tau^p} &= \|s\|_{L^p} + \|\nabla_B s\|_{L^p} + \|\nabla_B^2 s\|_{L^p} + \rho \|\nabla_\Sigma s\|_{L^p} \\ &\quad + \rho \|\nabla_B \nabla_\Sigma s\|_{L^p} + \tau \|\nabla_\Sigma^2 s\|_{L^p}. \end{aligned}$$

For the exact–gauge pair, with covariant Neumann boundary condition, use

$$\|\chi\|_{\mathcal{G}_\tau^p} = \|\chi\|_{W_\Sigma^{2,p}} + \rho^{-1} \|\nabla_B \chi\|_{W_\Sigma^{1,p}} + \rho^{-2} \|\nabla_B^2 \chi\|_{L^p}.$$

The Dolbeault component has its fixed $W^{1,p}$ graph norm. On the target leave the integrability, positive Cauchy–Riemann, and Coulomb rows unscaled, and multiply only the scalar HYM row by ρ^2 . These direct sum norms are denoted by $\mathcal{X}_{E,\tau}^p$ and $\mathcal{Y}_\tau^p$.

We next verify that these are norms on DFL’s first-order boundary problem, rather than on a formal second-order replacement. In DFL’s seam collar, write a spatial one-form as $b = q + \tau_0 ds$, where τ_0 is a field component. Their Hodge decomposition (5.17)–(5.21) is

$$(61) \quad q = h + d_\alpha \zeta + *_\Sigma d_\alpha \zeta'.$$

Here h is harmonic and

$$\zeta, \zeta' \in W_{s,\theta}^{1,p} W_\Sigma^{1,p} \cap L_{s,\theta}^p W_\Sigma^{2,p}.$$

The boundary conditions are

$$(62) \quad \tau_0|_{s=0} = 0, \quad \zeta'|_{s=0} = 0, \quad (h|_{s=0}, \nu|_{s=0}) \in V_\alpha,$$

where V_α is the harmonic Narasimhan–Seshadri graph. The exact potential ζ is unrestricted at the seam. The surface zero-form Laplacian has a uniform positive gap because every seam connection is irreducible. Hence (61) and its inverse are uniformly bounded in the displayed anisotropic spaces. This works with the full collar fields and requires no extension of a seam first jet.

Freeze the coefficients and take $d_\alpha^* d_\alpha e = \mu e$, $\mu > 0$. Substitution of

$$q = x\mu^{-1/2}d_\alpha e + y\mu^{-1/2}*_\Sigma d_\alpha e$$

in DFL (5.7) pairs (y, τ_0) and (x, φ) into two Cauchy–Riemann matrices. After the θ derivative is added and the system is squared, the off-diagonal first derivatives cancel. With the row normalization above, the two scalar symbols are

$$(63) \quad |\xi_B|^2 + \rho^2 \mu, \quad \rho^{-2} |\xi_B|^2 + \mu.$$

The first has the Dirichlet condition in (62); the second has the complementary Neumann condition supplied by the adjoint equation. They are respectively the metric and unitary-gauge blocks. Base Dirichlet Poincaré gives a gap for the first symbol, uniformly when $\rho^2 \mu$ tends to zero. The lower bound $\mu \geq \mu_0 > 0$ gives a gap for the second symbol, including its base-constant mode. Reflection and the multipliers in (63) give

$$(64) \quad \|s\|_{\mathcal{M}_\tau^p} \leq C \|L_{E,\tau}^{\text{Dir}} s\|_{L^p}, \quad \|\chi\|_{\mathcal{G}_\tau^p} \leq C \|G_{E,\tau}^{\text{Neu}} \chi\|_{L^p}.$$

The harmonic term in (61), coupled to ν by V_α , is the Dolbeault transmission block of Proposition 6.3. On the complement of its kernel, its smallest singular value is uniformly positive over C . DFL’s domain-straightening map in Proposition 5.27 acts separately on the harmonic, exact, and coexact entries of the full collar field. Its formula and inverse are uniformly bounded in the norms above. A partition of unity, coefficient freezing, and (64) therefore give

$$(65) \quad \|x\|_{\mathcal{X}_{E,\tau}^p} \leq C (\|\mathcal{D}_{E,\rho}^{\text{mix}} x\|_{\mathcal{Y}_\tau^p} + \|x\|_{L^p}).$$

The coefficient bounds needed to absorb the frozen errors follow from Theorem 6.5. The cross terms are triangular: holomorphic variation enters the HYM and Coulomb rows, whereas unitary gauge invariance and the Kähler identities prevent a gauge-to-HYM or metric-to-Coulomb term. Solving the harmonic row first and then the two rows in (64) gives uniformly bounded Schur coordinates.

It remains to remove the last term in (65). Otherwise there are $E_j \rightarrow E_\infty$, $\rho_j \rightarrow 0$, and reduced fields x_j of unit graph norm with $\mathcal{D}_{E_j,\rho_j}^{\text{mix}} x_j \rightarrow 0$. In the Schur coordinates, the harmonic component converges to its Dolbeault kernel projection. Orthogonality to the

smoothly varying kernel makes that projection zero. The two estimates in (64) then make the metric and gauge components tend to zero. This contradicts the unit graph norm. Thus

$$\|x\|_{\mathcal{X}_{E,\tau}^p} \leq C \|\mathcal{D}_{E,\rho}^{\text{mix}} x\|_{\mathcal{Y}_\tau^p}$$

on the reduced complement. Fixed-scale surjectivity from Proposition 6.3 and the open-mapping theorem give the right inverse and (60). $\square$

On the seam-admissible locus, $G_{\mathbb{R}}$ normalizes every gauge-side basepoint. A sequence whose basepoint approaches the seam is governed by the center-scale ratio in (25): finite ratio returns to the full-rank mixed chart, negative infinite ratio returns to the normalized gauge rank-drop chart, and positive infinite ratio is the ordinary holomorphic-sphere stratum treated in Proposition 3.2.

Proposition 6.8 (uniform attachment decay). *Let C be as in Theorem 6.7. In a common seam gauge near infinity there are constants C_k such that*

$$(66) \quad \sum_{j=0}^k |\zeta|^j |\nabla^j ((A_{E,\rho}, u_E) - (A_\infty, u_\infty))| \leq C_k |\zeta|$$

for $E \in C$ and $0 < \rho \leq \rho_0$, where $\zeta = 1/w$ is the removable coordinate at the attaching point.

Proof. Theorem 6.5 gives uniform smooth bounds up to the boundary point $\zeta = 0$. In temporal seam gauge, the value there is the Narasimhan–Seshadri representative of E_∞ . The positive classifying maps form a compact smooth family with the same value. Taylor’s theorem gives (66); the powers of $|\zeta|$ record the scale-invariant derivative norm. For ρ bounded away from zero the same estimate follows from ordinary compact parameter elliptic theory. Proposition 6.4 identifies these representatives with the clean cut-circle limits of an arbitrary degenerating sequence. $\square$

Theorem 6.9 (primitive master compactness). *Let (A_ν, u_ν) be mixed solutions with one matching concentration point of primitive charge and no ordinary instanton charge. Normalize the center by the energy barycenter, the scale by a fixed energy radius, and the remaining seam-preserving affine freedom. After passage to a subsequence, precisely one of the following occurs.*

- (1) *The coefficient pairs converge at finite seam ratio to a full-rank $E \in \mathfrak{M}_g^{\text{sa}}$. On every fixed rescaled core, the solutions converge in the graph norm of Theorem 6.7 to (A_{E,ρ_ν}, u_E) .*
- (2) *After negative affine recentering, they converge to a rank-drop $E \in \mathfrak{M}_g^{\text{sa}}$ whose basepoint lies in H_- . The same strong convergence holds to its Dirichlet reconstruction.*
- (3) *The seam ratio tends to positive infinity. The seam-centered component converges strongly to the constant mixed solution, while recentering at the escaping basepoint gives a unique simple minimal sphere. The convergence there is smooth away from the node.*

In every case the annulus from the primitive core to the residual solution has vanishing limiting energy. There is no further analytic scale or positive-energy component.

Proof. Apply finite-scale holomorphic doubling before taking a limit. On the rescaled discs this gives bundles

$$E_\nu \longrightarrow D_{R_\nu} \times \Sigma_g, \quad R_\nu \longrightarrow \infty,$$

framed on regular outer circles chosen from smooth convergence to the removable residual solution. The classifying loop there lies in a contractible fine-moduli chart and converges to

the attaching value. Cap it continuously in that chart. This cap is used only for relative degree; it is not asserted to be holomorphic. It may be chosen with symplectic area tending to zero.

Apply Lemma 4.2 on each expanding disc. The fixed-family identity (23) gives

$$d_\nu + \ell_\nu = 1.$$

The stable-reduction map is holomorphic on the disc, so its area is $d_\nu/2 + o(1)$, the error being the area of the shrinking cap. Gromov compactness followed by removal at infinity produces a stable map on $\mathbb{P}^1$, and the finitely many modification supports have a convergent subsequence. Positivity and the displayed identity leave either one degree-one component or one length-one modification. Muñoz's classification identifies the result with one primitive coefficient pair. The exact normal form (25) then gives the finite-ratio, negative-rank-drop, and positive-sphere alternatives. It also shows that a sphere and the limiting quasimap basepoint are two compactifications of the same unit of charge, not two components to be counted separately.

Consider the first two alternatives and fix a rescaled core K_R . For large ν , stable reduction supplies complex gauges in which the Dolbeault operators converge smoothly to that of E on K_R . The seam Narasimhan–Seshadri metrics converge smoothly because all seam fibers are stable and their trace-free zero-form Laplacians have a uniform spectral gap. Compare the original rescaled HYM metric with the reconstructed metric in these gauges. Their seam values agree in the limit. On the outer semicircle their difference tends to zero by residual convergence and Proposition 6.8.

Extend that outer boundary difference through a collar and subtract it. If e^{r_ν} is the remaining metric ratio, then r_ν has zero boundary value and satisfies an integrated linearized HYM equation

$$(67) \quad \left(\int_0^1 D_H \Phi(H_{\text{rec}} e^{tr_\nu}, \rho_\nu^2) dt \right) r_\nu = f_\nu, \quad \|f_\nu\|_{\mathcal{Y}_{\rho_\nu^2}^p(K_R)} \longrightarrow 0.$$

The maximum principle first gives $r_\nu \rightarrow 0$ in C^0 . The metric and gauge block estimates in (64) then give

$$\|r_\nu\|_{\mathcal{M}_{\rho_\nu^2}^p(K_R)} + \|\chi_\nu\|_{\mathcal{G}_{\rho_\nu^2}^p(K_R)} \longrightarrow 0,$$

where χ_ν is the unitary difference in the Coulomb slice. The harmonic part is precisely the convergent Dolbeault coefficient. The collar decomposition (61) therefore proves strong convergence in the full graph norm. The argument is uniform through gauge-side rank drop because the unstable fiber stays in the interior; the seam spectral gap and base Dirichlet estimate do not degenerate.

In the third alternative, the exact coefficient normal form converges on seam-centered compact sets to the constant stable bundle, so the preceding argument gives the constant mixed limit there. In the coordinate centered at the escaping basepoint the positive maps converge to

$$\eta \longmapsto [v\eta + w].$$

This is a simple minimal sphere. Its linearized operator is surjective by the regularity argument in Section 7.1; removal and elliptic bootstrapping give smooth convergence away from the node.

Finally let $R \rightarrow \infty$ along the good circles in Proposition 6.4. Its Chern–Simons identity makes the intervening energy tend to zero, and small-energy regularity upgrades this to first-order convergence on the selected annuli. The primitive charge has already been exhausted.

A further matching component or sphere would cost another half-unit, while an instanton would cost one unit. This proves exhaustion and the theorem. $\square$

6.6. The primitive gluing collar. Let $x_0 = (A_0, u_0)$ be a regular index-zero residual mixed solution and let

$$e_0 : \mathbb{R} \longrightarrow M_g$$

be its evaluation along the matching line. The primitive bubble stratum over x_0 is

$$(68) \quad \mathcal{B}_g(x_0) = (\mathbb{R}_{e_0} \times_{\text{ev}_\infty} \mathfrak{M}_g^{\text{sa}}) / G_{\mathbb{R}}.$$

When the fiber product is regular, the dimension calculation (38) gives

$$(69) \quad \dim_{\mathbb{R}} \mathcal{B}_g(x_0) = 3.$$

For $g \geq 3$ the generic fiber product is empty by (37); the construction below applies on every regular nonempty stratum before that final generic choice.

Theorem 6.10 (primitive mixed gluing). *Suppose x_0 and the fiber product in (68) are regular. For every compact $C \subset \mathcal{B}_g(x_0)$ there are $\rho_0 > 0$ and a smooth map*

$$(70) \quad \text{Glue} : C \times (0, \rho_0) \longrightarrow \mathcal{M}_\eta(\alpha, \beta)_4$$

with the following properties.

- (1) *As $\rho \rightarrow 0$, it converges in the Gromov–Uhlenbeck–stable-reduction topology to the corresponding residual-plus-bubble configuration.*
- (2) *It is a local diffeomorphism onto a neighborhood of the bubbled face.*
- (3) *Every mixed solution sufficiently close to C is represented uniquely by (70), modulo gauge and $G_{\mathbb{R}}$.*

Thus $\rho = 0$ is a collar face of the compactified index-four moduli space.

Proof. We give the construction with the estimates needed later. In the removable coordinate $\zeta = 1/w$, the bubble and residual solutions have the same stable flat seam value at $\zeta = 0$. Choose

$$(71) \quad R_\rho = \rho^{-1/2}, \quad r_\rho = \rho R_\rho = \rho^{1/2}, \quad T_\rho = \frac{1}{2} |\log \rho|.$$

Cut the bubble at $|w| = R_\rho$, cut the residual solution at physical radius r_ρ , and identify the annuli by $z = \rho w$. Use a common cutoff which is affine in logarithmic radius on a subcylinder of length comparable to T_ρ . Its cylindrical derivative is at most C/T_ρ . Use the same cutoff on both sides of the seam and the same flat representative. The matching condition is preserved exactly. The secondary perturbations vanish on this neighborhood and introduce no rescaled error.

The residual Taylor error is $O(r_\rho)$, the bubble-tail error is $O(R_\rho^{-1})$ by Proposition 6.8, and the cutoff commutator has L^p size $O(T_\rho^{-\kappa})$, where $\kappa = 1 - 1/p > 0$: its pointwise size is $O(T_\rho^{-1})$ and its transition cylinder has length $O(T_\rho)$. In the target norm of Theorem 6.7,

$$(72) \quad \|\mathcal{F}(x_{\text{pre}})\|_{\mathcal{Y}_{\rho^2}^p} \leq C(\rho^{1/2} + T_\rho^{-\kappa})$$

uniformly over C .

Let Q_0 be a right inverse for the regular index-zero residual operator and let $Q_{E,\rho}$ be the reduced inverse from Theorem 6.7. If K_0 and K_E are the two kernels, regularity of the evaluation fiber product is surjectivity of

$$\mathfrak{e} : \mathbb{R}_s \oplus K_0 \oplus K_E \longrightarrow TM_g, \quad \mathfrak{e}(\dot{s}, \xi_0, \xi_E) = d\text{ev}_0(\dot{s}, \xi_0) - d\text{ev}_\infty(\xi_E).$$

Choose a uniformly bounded right inverse for this finite-dimensional map over C . Cut a target field into residual, bubble, and neck pieces, apply Q_0 and $Q_{E,\rho}$, correct their harmonic attaching mismatch with that right inverse, and splice the source fields with the logarithmic cutoff. The harmonic neck modes are exactly those treated by $\mathfrak{e}$; irreducibility removes trace-free exact zero modes.

The resulting approximate inverse satisfies

$$\|D_\rho Q_\rho^{\text{app}} - \text{id}\| \leq C(T_\rho^{-1} + \rho^{1/2}).$$

The cutoff term is small because its logarithmic derivative tends to zero, the coefficient term is controlled by Proposition 6.8, and the collapsing block is controlled by Theorem 6.7. A Neumann series gives a right inverse Q_ρ with

$$(73) \quad \|Q_\rho\| \leq C.$$

The nonlinear term obeys the standard quadratic estimate

$$\|\mathcal{N}(\xi) - \mathcal{N}(\xi')\| \leq C(\|\xi\| + \|\xi'\|)\|\xi - \xi'\|.$$

The contraction equation

$$\xi = -Q_\rho(\mathcal{F}(x_{\text{pre}}) + \mathcal{N}(\xi))$$

therefore has a unique solution satisfying

$$\|\xi\| \leq C(\rho^{1/2} + T_\rho^{-\kappa}).$$

It varies smoothly in the bubble parameters and $\rho > 0$ and defines (70).

For the boundary smooth structure use

$$r = T_\rho^{-1} = \frac{2}{|\log \rho|}, \quad \rho = e^{-2/r}.$$

In this profile every positive power of ρ is flat at $r = 0$, and the cutoff is fixed after dividing the neck coordinate by T_ρ . The parameter-dependent contraction theorem makes the corrected family smooth for $r \geq 0$. Increasing r has the same inward orientation as increasing ρ .

For uniqueness, fix center and scale by the energy barycenter and the half-energy radius and impose Coulomb gauge. The derivative of these two conditions on translation and positive dilation is nonsingular. Two glued solutions with the same limit differ in the reduced complement; (73) and the quadratic estimate force the difference to vanish.

Conversely, Theorem 6.9 assigns every nearby solution a residual solution, primitive coefficient, center, and scale, and makes it converge to the corresponding pregluing in the source graph norm on both cores and the complete neck. It therefore lies in the same contraction ball as the correction constructed above. Uniqueness of the fixed point proves local surjectivity without assuming uniqueness of an abstract stable-reduction limit.

Changing cutoffs, gauges, or exponential charts changes the pregluing by a term tending to zero in the source norm. The two corrections solve the same equation in the same center-scale slice, so contraction uniqueness identifies them. Coefficient-chart overlaps are smooth by Dirichlet reconstruction, including at gauge rank drop. At positive infinite ratio, the exact coefficient normal form is the ordinary nodal-sphere smoothing; standard simple-sphere gluing gives the compatible raw compactification chart. This proves the overlap assertion and completes the collar theorem. $\square$

A rank-drop basepoint on the gauge side produces no additional end: $G_{\mathbb{R}}$ normalizes it, so it belongs to the same Dirichlet family as the full-rank bubbles. A positive-side rank-drop limit is instead a constant mixed component with an ordinary minimal sphere. Its

standard nodal sphere collar is part of the raw compactification and is compatible with the finite seam-ratio chart through (25); Proposition 3.2 shows that the point incidence misses it. Thus Theorem 6.10 is required only on the full-rank and gauge rank-drop colors of the primitive matching face.

7. TRANSVERSALITY, INCIDENCE, AND THE BOUNDARY FORMULA

7.1. Compact stratification. Through Fredholm index four, the compactified mixed moduli space has the following strata:

- (1) the unbroken, unbubbled mixed stratum;
- (2) ordinary gauge and symplectic Floer breakings;
- (3) interior sphere and outer-boundary disc strata, which have codimension at least two in the point cut by Proposition 3.2;
- (4) the primitive matching face constructed in Theorem 6.10;
- (5) the full-rank and Segre rank strata inside that face.

There are no further scale-tree strata. Theorem 3.3 and Corollary 3.4 give one charge-one atom; Proposition 4.3 gives each individual doubled family's algebraic type as a minimal line or one rank-drop modification; Proposition 6.4 excludes hidden analytic necks; and Theorem 6.10 constructs a collar of the resulting face.

The obstruction to extending DFL's perturbation construction verbatim from index three to index four is now visible. Their local cokernel-separation argument is index independent, but their finite-cover argument needs compactness of the nonregular locus. At index four that compactness becomes available only after the primitive face has been added.

The ordinary bubbling strata cause no further transversality problem. Index exhaustion makes every sphere or disc component minimal and hence simple. Away from the matching neighborhood, the usual universal almost-complex-structure argument regularizes these components and their nodal evaluation maps. In the region where the complex structure is the fixed structure J_* , a minimal sphere is one of the extension lines in M_g . For its bundle E on $\mathbb{P}^1 \times \Sigma_g$, stability of the fibers gives $R^0 \pi_* \text{End}_0 E = 0$ and identifies $R^1 \pi_* \text{End}_0 E$ with the pullback of TM_g . The Leray sequence and the vanishing in Proposition 6.3 therefore give $H^1(\mathbb{P}^1, f^* TM_g) = 0$, which is surjectivity of the sphere operator. A minimal outer-boundary disc is regular for a generic allowed domain-dependent almost complex structure. The standard simple-curve gluing construction then gives neighborhoods of these codimension-two strata; for the sphere case see [MS04, Chapter 10], and for the monotone Lagrangian boundary case see [Oh93]. These ordinary neighborhoods are used below only to compactify the transversality problem; Proposition 3.2 shows that the point cut misses them.

7.2. Evaluation transversality. Let $\mathcal{P}$ be the DFL space of standard secondary perturbations: holonomy perturbations on the gauge side and domain-dependent almost-complex variations on the symplectic side, supported away from the matching neighborhood.

Let $\mathcal{S}$ be a stratum in an algebraic Whitney stratification for which i_g and the restrictions of ev_∞ to the full-rank and Segre strata are stratified submersions. On the universal residual moduli space consider

$$(74) \quad \text{Ev}_\gamma : \mathcal{M}_0^{\text{univ}}(\alpha, \beta) \times \mathbb{R}_s \longrightarrow M_g, \quad \text{Ev}_\gamma([A, u], \eta, s) = [A|_{(s,0) \times \Sigma_g}].$$

The matching condition gives the same value from u .

Proposition 7.1 (relative seam evaluation). *After an arbitrarily small finite-dimensional enlargement of $\mathcal{P}$, the map (74) is transverse to every stratum $\mathcal{S}$ occurring in the compact*

energy range. The enlargement is supported away from the matching neighborhood and can be made relative to all previously regular moduli spaces.

Proof. Write $\mathcal{F}_\eta$ for the mixed equation and fix a point at which Ev_γ lies in $\mathcal{S}$. Transversality of $(\mathcal{F}, \text{Ev}_\gamma)$ to the zero section times $\mathcal{S}$ is equivalent to surjectivity of

$$(75) \quad (\xi, \dot{\eta}, \dot{s}) \longmapsto (D_{(A,u)}\xi + P_{(A,u)}\dot{\eta}, \pi_{N\mathcal{S}} d\text{Ev}_\gamma(\xi, \dot{s})).$$

Suppose a pair (λ, ν) annihilates its image. Here λ is an adjoint mixed field and $\nu \in N^*\mathcal{S}$. Variation in $\dot{\eta}$ shows that λ annihilates every allowed holonomy and almost-complex direction. On the complement of the evaluation point it satisfies DFL's homogeneous adjoint equation. If its gauge component is nonzero on the allowed support, the holonomy direction constructed in the proof of [DFL26, Proposition 3.23] pairs nontrivially with it. If the gauge component vanishes but the symplectic component is nonzero, DFL's variation of the almost complex structure at a point where du is nonzero does the same. Their constant-map argument covers the remaining case. Thus λ vanishes on the perturbation supports. Unique continuation on both pieces makes it vanish away from the evaluation point.

The weak adjoint equation is now

$$(76) \quad D_{(A,u)}^*\lambda + \text{Ev}_\gamma^*\nu = 0.$$

The adjoint field belongs to the ordinary dual Sobolev space, so vanishing off one point means that $\lambda = 0$ as an L^q field. Equation (76) reduces to $\text{Ev}_\gamma^*\nu = 0$. Smooth configuration variations realize an arbitrary tangent vector in TM_g at the seam trace: prescribe the matching trace locally and extend it to the two configuration spaces. Hence $\nu = 0$, proving surjectivity of (75).

The relevant universal zero set is compact after the ordinary and primitive strata have been added. Finitely many of the separating directions above therefore suffice. Sard–Smale applied to this finite-dimensional family gives the assertion. The usual descending smallness choice makes the construction relative to earlier regular spaces and preserves DFL's nonnegative-energy bound. $\square$

Choose a Whitney stratification of the algebraic image of i_g and apply Proposition 7.1 simultaneously to its finitely many strata in the compact energy range. For $g \geq 3$, the residual path misses the image by (37). For $g = 2$, it avoids the branch locus and meets the regular four-sheeted locus. When $g = 1$, the target is a point; the excess Jacobian is regularized by the universal incidence section rather than by evaluation.

Proposition 7.2 (relative index-four regularity). *The secondary perturbations can be chosen so that all mixed moduli spaces needed by the point operation are regular through index four, including their residual, broken, primitive, and rank strata. The perturbations remain identically zero in the fixed matching neighborhood.*

Proof. First choose the DFL perturbations so that all lower-index residual and broken strata are regular. Proposition 7.1 makes the primitive evaluation fiber products regular. Fixed-scale regularity and Theorem 6.10 then give parameterized collars of the primitive faces. Ordinary breaking collars are regular by DFL Floer gluing, and the simple sphere and disc collars are regular by the preceding ordinary-bubble argument.

These constructions also apply to a sufficiently small family of secondary perturbations: the right inverses and gluing maps vary continuously, so regularity of the compact union of boundary strata is an open condition. Remove smaller collar neighborhoods from the parameterized compactified index-four space. Properness of DFL compactness together with

Theorem 6.10 makes the remaining universal core compact. At each nonregular solution in this core, DFL’s local cokernel-separation argument supplies finitely many perturbation directions that kill its cokernel. A finite subcover and Sard’s theorem, exactly as in the proof of [DFL26, Proposition 3.23], give one arbitrarily small parameter for which the core is regular. Smallness preserves the already regular collar strata. All perturbation directions are supported away from the seam, so the local charge, stable-reduction, Dirichlet, and primitive gluing models are unchanged. $\square$

7.3. Extension of the point incidence. Blow up the pair $(z - z_0, \rho)$ near a primitive matching point. The front face has affine coordinate $w = (z - z_0)/\rho$. In the common flat seam gauge on a good-circle annulus, let g_0 and g_E be the framed transition functions of the residual and reconstructed bubble bundles. For $\rho > 0$, splice $g_0^{-1}g_E$ with the logarithmic cutoff used in the linear gluing construction. At $\rho = 0$, retain the residual cocycle on the principal component and the bubble cocycle on the front face, identified at $w = \infty$. No-neck convergence makes the cocycles converge on the overlap. This constructs the universal projective bundle on the gluing blowup.

The fine-moduli ambiguity tensors a rank-two universal bundle by a line from the parameter space; its adjoint is unchanged. We therefore obtain a canonical adjoint bundle over the compactified chart, up to an isomorphism fixed at the interpolation ends. It is smooth on every stratum and through gauge rank drop by the uniform reconstruction theorem. On the positive infinite-ratio chart it is the standard universal bundle over the ordinary stable-map sphere collar. Its front-face clutching class is the class used in Theorem 3.3, so there is no additional neck class.

Two sections of the extended bundle define a stratified codimension-four linear-dependence cycle on

$$\overline{\mathcal{M}}_\eta(\alpha, \beta)_4 \times \mathbb{R}_t.$$

Proposition 7.3 (relative point-incidence transversality). *The two interpolating sections can be chosen, relative to the fixed transverse representatives at $t = \pm\infty$, so that their rank-one degeneracy locus is transverse to every stratum of the index-three and compactified index-four universal evaluation maps. The rank-zero locus is empty. On the genus-one primitive stratum the induced normal section of the excess determinant line is transverse to zero.*

Proof. At a point of the rank-one stratum, independent pointwise variations of the two sections surject onto the normal space of the determinantal variety in $\text{Hom}(\mathbb{C}^2, \mathbb{C}^3)$. The same statement, with a larger normal space, holds at the rank-zero stratum. Section variations supported in an arbitrarily small neighborhood of a given evaluation point realize these pointwise variations. Relative Thom transversality, applied successively to the finite Whitney stratification from Proposition 7.2, therefore gives simultaneous transversality while leaving collars of $t = \pm\infty$ fixed. The rank-zero stratum has real codimension twelve, greater than the dimension of every evaluation stratum here, and is consequently empty. On the genus-one excess family, prescribe the first complex normal derivative as a section of the rank-two coefficient bundle. Its real rank is four while the base has real dimension two, so relative transversality makes it nowhere zero. Project the second normal derivative to the resulting complex cokernel line. A further relative variation makes that section transverse to zero. These are the two derivatives used in Proposition 7.6. $\square$

Near a primitive face, (69) and the interpolation parameter give dimension four. Proposition 7.3 cuts the face in an oriented zero-manifold and its gluing collar in a one-manifold.

Together with Proposition 3.2 and Proposition 7.2, this gives the required compactness statement.

Proposition 7.4 (compact point-cut moduli spaces). *For compatible generic choices, $\mathcal{L}_0^q(\alpha, \beta)$ is an oriented zero-manifold and $\overline{\mathcal{L}}_1^q(\alpha, \beta)$ is a compact oriented one-manifold. Its finite- t boundary consists only of ordinary Floer breakings and primitive matching configurations.*

Proof. Propositions 7.2 and 7.3 give the asserted dimensions and orientations on the top stratum and on each compactification stratum. The primitive gluing collar makes the primitive cut compact, while Proposition 3.2 excludes the ordinary sphere and disc faces. DFL compactness, index-four exhaustion, and primitive stable reduction leave only the boundary types listed in the statement. The index-three cut is bubble-free by Proposition 3.1; its zero-dimensional compactification therefore has no additional ends. $\square$

7.4. Determinant locality. Orient $G_{\mathbb{R}}$ by translation followed by positive dilation, orient the matching-center and interpolation lines $\mathbb{R}_s, \mathbb{R}_t$ by increasing coordinate, and write $\mathfrak{g}_{\mathbb{R}}$ for the oriented Lie algebra of $G_{\mathbb{R}}$. Let $\mathcal{N}_{p_1}$ be the complex rank-two normal bundle of the rank-one determinantal stratum in $\text{Hom}(\mathbb{C}^2, \mathbb{C}^3)$, with DFL's positive p_1 coorientation.

Proposition 7.5 (determinant excision). *At a regular primitive configuration (x_0, E) , the determinant line of the glued index-four operator has a natural excision isomorphism*

$$(77) \quad \det \mathcal{D}_{g_1} \cong \det \mathcal{D}_{x_0} \otimes \det \mathcal{D}_{E, \rho}^{\text{mix}} \otimes \det \mathbb{R}_s \otimes (\det TM_g)^* \otimes (\det \mathfrak{g}_{\mathbb{R}})^* \otimes \mathbb{R}_{\rho}.$$

Here $\mathbb{R}_{\rho}$ is the gluing-normal line, oriented by increasing ρ . The isomorphism is continuous across the full-rank and rank-drop strata. On the boundary point cut, its finite-dimensional local factor is

$$(78) \quad \begin{aligned} \mathfrak{o}_g(E) = & \det \mathcal{D}_{E, \rho}^{\text{mix}} \otimes \det \mathbb{R}_s \otimes (\det TM_g)^* \otimes (\det \mathfrak{g}_{\mathbb{R}})^* \\ & \otimes \det \mathbb{R}_t \otimes (\det \mathcal{N}_{p_1})^*. \end{aligned}$$

Thus the orientation line of a primitive point-cut boundary point is $\det \mathcal{D}_{x_0} \otimes \mathfrak{o}_g(E)$.

Proof. The patched inverse in Theorem 6.10 deforms the linearized glued operator, through Fredholm operators, to the direct sum of the residual and fixed-domain bubble operators together with the finite-dimensional matching map. At the level of kernels that map is

$$(79) \quad \mathbb{R}_s \oplus H^1(\mathbb{P}^1 \times \Sigma_g, \text{End}_0 E) \longrightarrow TM_g, \quad (\dot{s}, \dot{E}) \longmapsto de_0(\dot{s}) - d\text{ev}_{\infty}(\dot{E}).$$

It is onto precisely when the evaluation fiber product (68) is regular. Proposition 7.1 provides this condition after the generic relative choice of perturbations. Its kernel contains the infinitesimal $G_{\mathbb{R}}$ orbit, and the tangent of the primitive face is the quotient by that orbit. The determinant functor applied to the two resulting short exact sequences gives all factors in (77). The gluing parameter supplies $\mathbb{R}_{\rho}$.

Proposition 6.3 identifies the bubble determinant with the complex determinant of the Dolbeault complex and shows that its obstruction space vanishes. The matching target TM_g has its complex orientation. The constructions depend continuously on E , including on the Segre locus, because the patched inverses and the Dolbeault family do. Finally, adding $\mathbb{R}_t$ and applying the determinant functor to the surjective normal derivative of the point incidence contracts $\det \mathcal{N}_{p_1}$ and gives (78). $\square$

All factors in $\mathfrak{o}_g(E)$ are supported in the standard surface-seam neighborhood. They are independent of the two three-manifolds and of the residual operator; the latter contributes

only $\det \mathcal{D}_{x_0}$, with the orientation used in the comparison count N . Consequently the primitive contribution is multiplication of the comparison count by a universal integer

$$(80) \quad c_g^{\text{loc}} \in \mathbb{Z}$$

depending only on the genus, the integral p_1 convention, and DFL’s seam orientation. Summing over index-zero residual solutions gives

$$(81) \quad B_q = c_g^{\text{loc}} N.$$

Proposition 7.6 (algebraic local determinant). *With the complex orientations in Proposition 6.3, the absolute local count is four in genus two. In genus one the excess local determinant line over $J(\Sigma_1)$ is canonically*

$$(82) \quad \det \mathcal{E}_\zeta.$$

Hence its absolute local count is eight.

Proof. The Dolbeault reduction makes the finite-dimensional normal problem complex linear. In genus two, evaluation at infinity is the holomorphic map $i_2 : N_2 \rightarrow M_2$. Over a regular value, the remaining normalized affine, center, and interpolation variables form the standard complex rank-two normal slice to the two-section determinantal cycle. To see its relative degree, fix an extension basis (v, w) with v representing the attaching line. A nearby pencil is

$$[vz_0 + (av + cw)z_1].$$

The complex number c is the Segre smoothing coordinate. The matching center and normalized interpolation displacement form a second complex coordinate ζ ; the real affine quotient removes the complementary translation and positive-dilation directions. Before applying relative transversality on the complement, prescribe the two-section pair on a tubular neighborhood of this slice to be the Thom model whose incidence-normal map is

$$(c, \zeta) \longmapsto (c, \zeta)$$

on every small normal disk. It is nonzero on the disk boundary and has one transverse zero of complex multiplicity one. The pair extends over the complement by Proposition 7.3. Any two such Thom choices are homotopic through nonzero boundary maps and have the same degree. The ordinary positive-sphere chart has negative expected incidence dimension, while the map extends transversely across the real-codimension-two gauge rank-drop stratum, so no degree escapes through the compactification. Thus every point of $i_2^{-1}(y)$ has complex multiplicity one. Proposition 5.2 gives four such points counted with multiplicity.

For $g = 1$, the character variety is a point and every projective coefficient pair has rank one. After the basepoint and the real affine orbit are normalized, the remaining excess parameter is $L \in J(\Sigma_1)$. Applying the determinant functor to the Dolbeault normal complex and to the rank-one incidence normal sequence cancels the fixed domain, affine, and Dirichlet metric factors. More explicitly, the normal incidence complex over L is

$$(83) \quad \mathbb{C}_{\text{norm}} \longrightarrow H^0(\mathbb{P}^1, \mathcal{O}(1)) \otimes (\mathcal{E}_1)_L.$$

The source is the trivial complex line obtained from the center and interpolation normal after the affine quotient; the target records the two extension coefficients. Use the relative freedom in the first normal derivative of the incidence pair to make this arrow a nowhere-zero smooth bundle map. Its cokernel is then a complex line bundle Q fitting into

$$0 \longrightarrow \mathbb{C}_{\text{norm}} \longrightarrow \mathcal{E}_\zeta \longrightarrow Q \longrightarrow 0.$$

The remaining normal derivative of the incidence pair projects to a section σ_Q of Q . A primitive configuration lies in the point cut exactly when σ_Q vanishes, and Proposition 7.3 makes this section transverse to zero. Since the source line is canonically trivial,

$$Q \cong \det \mathcal{E}_\zeta,$$

which is (82). This calculation is unaffected by the metric block because that block is an invertible family with its canonical determinant trivialization. Equation (44) gives

$$\langle c_1(\det \mathcal{E}_\zeta), [J(\Sigma_1)] \rangle = 8.$$

The complex zero-set orientation gives the asserted absolute counts. $\square$

7.5. Evaluation of the local coefficient. For $g \geq 3$, Proposition 7.1 and (37) make the attachment fiber product empty. Hence

$$(84) \quad c_g^{\text{loc}} = 0, \quad g \geq 3.$$

Proposition 7.6 gives absolute local counts four and eight in genera two and one, respectively.

It remains to compare these complex orientations with DFL's coherent mixed orientation. The excision isomorphism (77) proves that this comparison is the same universal sign on every target. Evaluate it on the cylinder splitting of $\Sigma_g \times S^1$. DFL orient their index- $8k + 4$ gauge path by gluing the positive generator sphere to a constant strip and require the symplectic orientations to agree with that sphere-gluing map [DFL26, Remark 4.76]. Independently, Proposition 5.3 gives $+8N$ and $+4N$ from the Floer and quantum ring relations. Since N is an isomorphism, the local complex orientations agree with the DFL boundary orientation. Therefore

$$(85) \quad c_g^{\text{loc}} = \begin{cases} 8, & g = 1, \\ 4, & g = 2, \\ 0, & g \geq 3. \end{cases}$$

This calibration is not circular: it uses surface compatibility, not point compatibility.

7.6. The oriented boundary count.

Theorem 7.7 (point-class comparison). *Let $(Y_\#, E_\#)$ have an admissible splitting $Y_\# = Y \cup_\Sigma (-Y')$ in the embedded-Lagrangian DFL setting. Assume the flat connections are irreducible and use compatible standard perturbations that vanish near the matching line. Let $q \in \Sigma_j$ and write g_j for the genus of that component. Over $\mathbb{Q}$, the comparison chain map and the positive quarter-Pontryagin point operations satisfy*

$$(86) \quad Nm_q^G - m_q^S N = d_S K_q + K_q d_G + \kappa_{g_j} N, \quad \kappa_g = \begin{cases} 2, & g = 1, \\ 1, & g = 2, \\ 0, & g \geq 3. \end{cases}$$

Proof. The zero-dimensional index-three point cut defines K_q by (8); it is bubble-free by Proposition 3.1. Proposition 7.4 gives a compact oriented one-manifold at index four.

Its ordinary broken boundary points contribute $d_S K_q + K_q d_G$. Escape of the interpolation parameter to the gauge and symplectic ends contributes $Nm_q^G - m_q^S N$ with DFL's end convention; this is the same orientation calculation as [DFL26, Proposition 7.12]. By Proposition 3.2, no finite- t interior sphere or outer disc contributes. Corollary 3.4, Proposition 4.3,

and Theorem 6.10 show that the only remaining boundary is the primitive matching face. Determinant locality and (85) make its integral contribution $c_{g_j}^{\text{loc}} N$.

The signed boundary count of a compact oriented one-manifold is zero. Using the integral identity (10) and dividing by four gives (86). $\square$

The same construction with one additional continuation parameter proves independence of the interpolation sections and secondary perturbations. The parameterized compactification has the same primitive faces; the local coefficient is unchanged by determinant locality, and its remaining boundary gives the standard chain homotopy between choices.

8. INTEGRAL NORMALIZATION AND CONSEQUENCES

Theorem 8.1 (integral point cycle). *Before division by four, the operations represented by the positive integral degeneracy cycle satisfy*

$$(87) \quad Nb_q^G - b_q^S N = d_S K_q^{p_1} + K_q^{p_1} d_G + c_{g_j} N, \quad c_g = \begin{cases} 8, & g = 1, \\ 4, & g = 2, \\ 0, & g \geq 3. \end{cases}$$

Proof. This is the oriented boundary count in the proof of Theorem 7.7 before the definitions $m_q^G = \frac{1}{4}b_q^G$, $m_q^S = \frac{1}{4}b_q^S$, and $K_q = \frac{1}{4}K_q^{p_1}$ are applied. $\square$

Corollary 8.2. *Define*

$$\widehat{m}_q^S = m_q^S + \kappa_{g_j} \text{id}.$$

Then N intertwines m_q^G and $\widehat{m}_q^S$ up to chain homotopy:

$$(88) \quad Nm_q^G - \widehat{m}_q^S N = d_S K_q + K_q d_G.$$

In particular, the induced Atiyah–Floer isomorphism intertwines these corrected point operations on homology.

Proof. Move $\kappa_{g_j} N$ to the left-hand side of (86) and use the definition of $\widehat{m}_q^S$. $\square$

8.1. The full corrected operation algebra. Theorem 7.7 completes the list of generator comparisons. We record the resulting algebra statement, first in the integral normalization used by the degeneracy-cycle constructions and then in the quarter-Pontryagin normalization of the introduction.

Give $H_k(\Sigma; \mathbb{Z})$ degree $4 - k$. For a commutative ring R , let $V_R = H_*(\Sigma; \mathbb{Z}) \otimes_{\mathbb{Z}} R$ and define

$$(89) \quad \mathbb{A}_{p_1}(\Sigma; R) = T_R(V_R) / \left\langle x \otimes y - (-1)^{|x||y|} y \otimes x \right\rangle =: \text{Sym}_R^{\text{gr}}(V_R)$$

for homogeneous x, y . This presentation fixes the meaning of “graded-commutative” over rings with two-torsion. In particular, it imposes $2x^2 = 0$, but not $x^2 = 0$, for an odd generator. If 2 is invertible in R , then

$$(90) \quad \mathbb{A}_{p_1}(\Sigma; R) \cong \text{Sym}_R(H_0(\Sigma; R) \oplus H_2(\Sigma; R)) \otimes_R \Lambda_R H_1(\Sigma; R).$$

Lemma 8.3 (integral operation relations). *On gauge and symplectic Floer homology with coefficients in R , the integral universal-class operations satisfy*

$$b_x b_y = (-1)^{|x||y|} b_y b_x$$

for homogeneous $x, y \in V_R$. Hence the homology actions factor through (89). For an odd loop class this asserts $2b_x^2 = 0$; no vanishing of b_x^2 in characteristic two is used.

Proof. Work first over $\mathbb{Z}$. The operation b_x is the cap operation associated with the slant class $p_1(\mathbb{V})/x$ on the relevant configuration space; DFL construct its gauge and symplectic representatives from the same oriented integral degeneracy locus in Section 7.2 of [DFL26]. For two classes, use trajectories with two ordered marked cutdowns. Stretching the interval between the marked points gives the composite $b_x b_y$. Bringing the points together gives the cutdown by the cup product

$$(p_1(\mathbb{V})/x) \smile (p_1(\mathbb{V})/y).$$

The compactified one-parameter family has only its two endpoint cutdowns and Floer breakings; the latter are the differential terms in the resulting chain homotopy. This is the usual cap-module proof on the gauge side and the quantum-module proof on the symplectic side, with sphere terms included in the quantum product. Interchanging the two factors in the coincident cutdown changes its orientation by $(-1)^{|x||y|}$. Reversing the deformation therefore proves the displayed relation on homology.

All parameter spaces, degeneracy cycles, and determinant orientations in this argument are integral. Tensoring the resulting chain homotopies with R proves the assertion over an arbitrary commutative ring. Taking $x = y$ odd gives exactly $2b_x^2 = 0$. $\square$

For $\sigma \in H_k(\Sigma; \mathbb{Z})$, let b_σ^G and b_σ^S denote the gauge and symplectic operations obtained from the positive integral degeneracy cycle $p_1/[\sigma]$. The constructions are extended R -linearly after changing coefficients. Surface and loop operations are left unchanged. For $q \in \Sigma_j$, set

$$(91) \quad \widehat{b}_q^S = b_q^S + c_{g_j} \text{id}, \quad c_g = \begin{cases} 8, & g = 1, \\ 4, & g = 2, \\ 0, & g \geq 3. \end{cases}$$

The resulting homology actions are denoted by ρ_{G,p_1}^R and $\widehat{\rho}_{S,p_1}^R$, and the scalar extension of N is denoted by N_R .

Corollary 8.4 (full corrected operation algebra). *Under the hypotheses of Theorem 7.7, for every commutative ring R the base-changed comparison isomorphism is an isomorphism of $\mathbb{A}_{p_1}(\Sigma; R)$ -modules:*

$$(92) \quad N_R \rho_{G,p_1}^R(a) = \widehat{\rho}_{S,p_1}^R(a) N_R, \quad a \in \mathbb{A}_{p_1}(\Sigma; R).$$

For each chosen word representing a , the equality is induced by an explicit chain homotopy. No strict chain-level graded-commutativity or homotopy-coherence assertion is included.

Proof. Work first over $\mathbb{Z}$. For a homogeneous generator x , write A_x for the gauge operation and B_x for the corrected symplectic operation. The detailed DFL surface and loop cutdowns use the integral degeneracy locus representing p_1 . Their comparison theorem therefore supplies maps K_x for $x \in H_1(\Sigma) \oplus H_2(\Sigma)$ such that

$$(93) \quad N A_x - B_x N = d_S K_x + (-1)^{|x|} K_x d_G.$$

For $x \in H_0(\Sigma)$, Theorem 8.1 gives the same identity after the correction in (91) is moved to the symplectic side.

Suppose (93) has been extended to homogeneous words u and v . Define

$$(94) \quad K_{uv} = K_u A_v + (-1)^{|u|} B_u K_v.$$

Then

$$\begin{aligned} NA_u A_v - B_u B_v N &= (NA_u - B_u N)A_v + B_u(NA_v - B_v N) \\ &= d_S K_{uv} + (-1)^{|u|+|v|} K_{uv} d_G. \end{aligned}$$

The second equality is the graded Leibniz rule, using that A_v and B_u are chain maps of degrees $|v|$ and $|u|$. Induction proves the identity for every word, and linearity treats every polynomial expression. Passing to homology proves (92) for every ordered word. Lemma 8.3 shows that the resulting homology operators factor through (89); the construction of K_a may depend on the chosen word representing a .

DFL prove that N is an isomorphism of chain complexes over $\mathbb{Z}$. Tensoring N , its inverse, the operation maps, and the displayed homotopies with an arbitrary commutative ring R preserves all identities. No flatness hypothesis on R is required. $\square$

Corollary 8.5 (quarter-Pontryagin normalization). *Suppose 4 is invertible in R and put $m_\sigma = \frac{1}{4}b_\sigma$. Then Corollary 8.4 holds for the normalized observable action, with the corrected point class*

$$(95) \quad \widehat{m}_q^S = m_q^S + \kappa_{g_j} \text{id}, \quad (\kappa_1, \kappa_2, \kappa_{g \geq 3}) = (2, 1, 0).$$

In particular, over $\mathbb{Q}$ the DFL comparison intertwines the full corrected Donaldson observable algebra

$$\text{Sym}(H_0(\Sigma; \mathbb{Q}) \oplus H_2(\Sigma; \mathbb{Q})) \otimes \Lambda H_1(\Sigma; \mathbb{Q}).$$

Proof. Divide each integral generator identity and its homotopy by four. The point coefficients in (91) become respectively 2, 1, and 0. $\square$

Corollary 8.6 (operation images and kernels). *For every coefficient ring covered by Corollary 8.4, conjugation by N_R identifies the two image operation algebras. Moreover,*

$$(96) \quad \ker \rho_{G,p_1}^R = \ker \widehat{\rho}_{S,p_1}^R,$$

so the identity on $\mathbb{A}_{p_1}(\Sigma; R)$ induces an isomorphism of the two operation quotients.

Proof. Equation (92) and invertibility of N_R give (96); the same conjugacy identifies the images. $\square$

Remark 8.7. Corollary 8.4 concerns the action of the full observable algebra. It does not say that the mixed comparison map preserves a gauge-theoretic pair-of-pants product, and the wordwise homotopies (94) do not constitute an A_∞ module morphism. An integral operation representing $\frac{1}{4}p_1/[\sigma]$ still requires the divisibility data discussed in Remark 8.9; the all-coefficient statement instead uses the integral p_1 cycle.

Remark 8.8. The correction in genera one and two cannot be removed by changing the compactification. Proposition 5.3 detects it algebraically on $\Sigma_g \times S^1$. It is therefore part of the point operation itself under the DFL comparison, rather than an artifact of the proof.

Remark 8.9. The quarter-Pontryagin statement is asserted over $\mathbb{Q}$. An integral operation representing $\frac{1}{4}p_1/[q]$ requires a divisibility choice for the universal bundle that is not used here. The integral theorem is Theorem 8.1, formulated directly with the degeneracy cycle $p_1/[q]$.

AVAILABILITY

The manuscript source is available at

<https://github.com/bwuebben/donaldson-atiyah-floer>.

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